MIREIA RIBERA | ACCESSIBILITAT DIGITAL
| EXPERIÈNCIA D'USUARI
|VISUALITZACIÓ DE DADES

Presentation and Visualization.



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Graphic design



Graphic design *for what?*

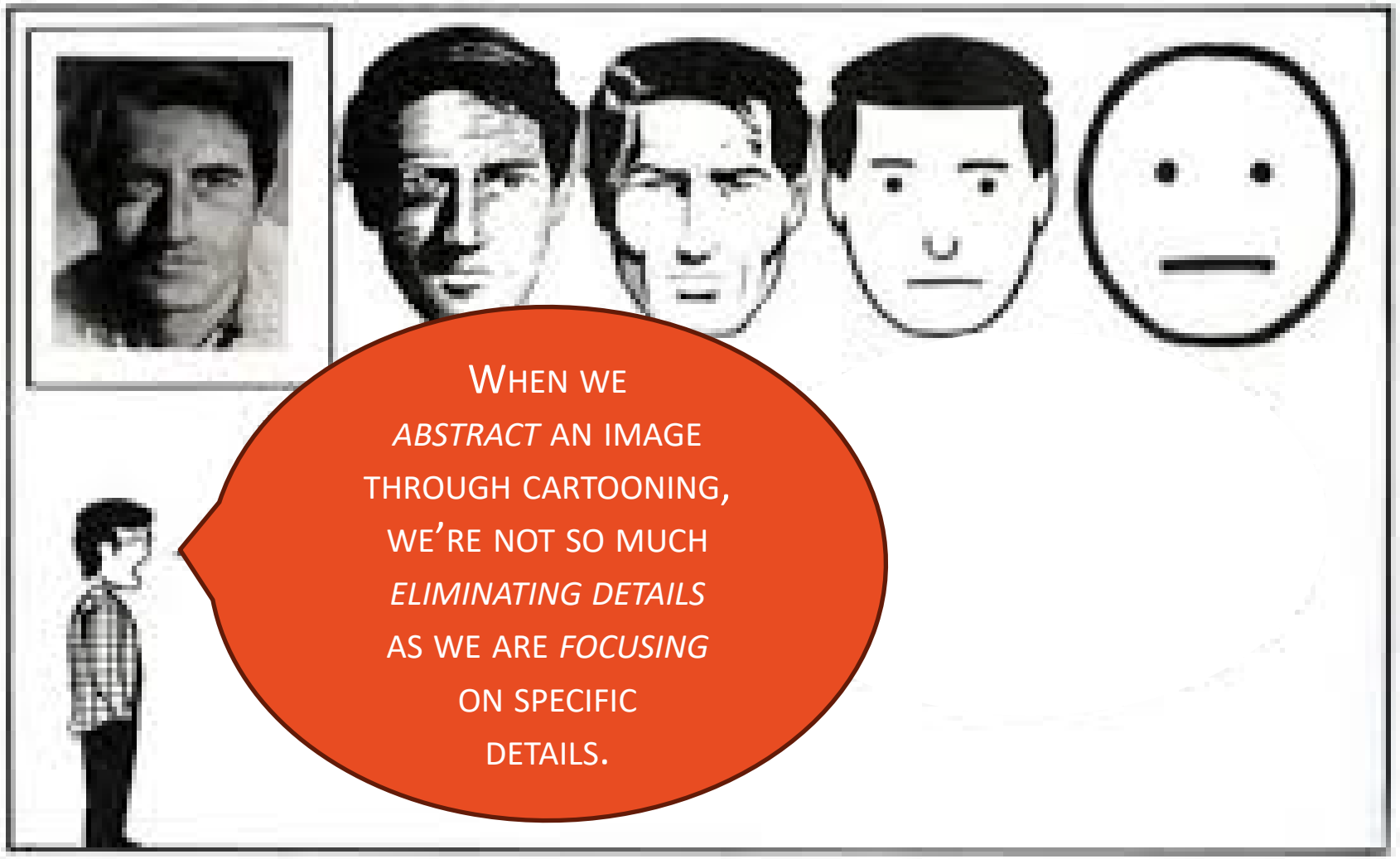
Visualization lets us use

our human perceptual capabilities

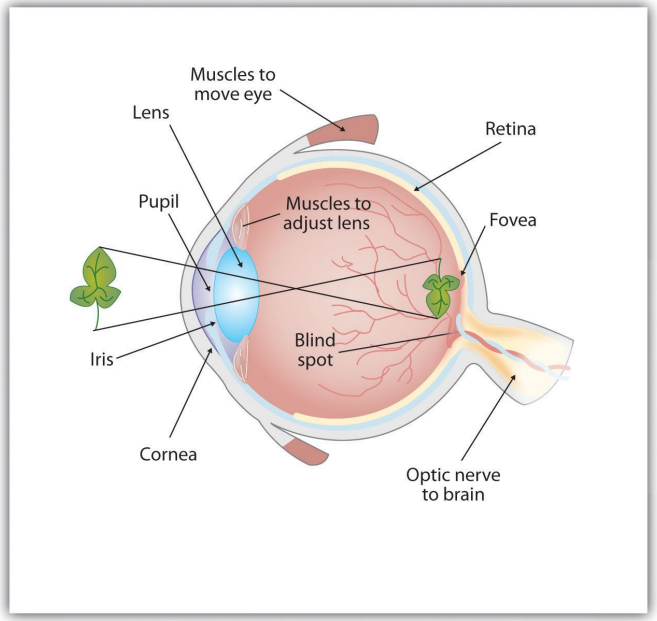
to ***gain insights***

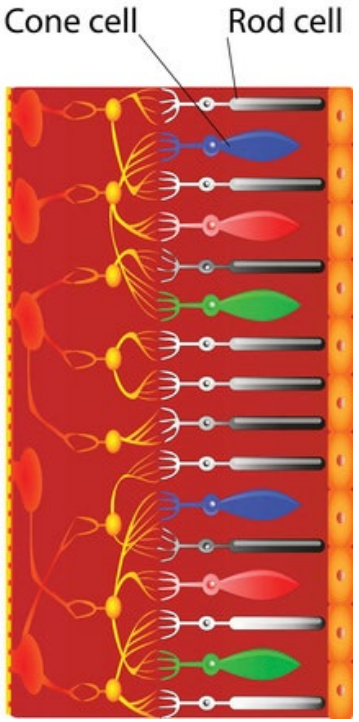
into abstract data sets.

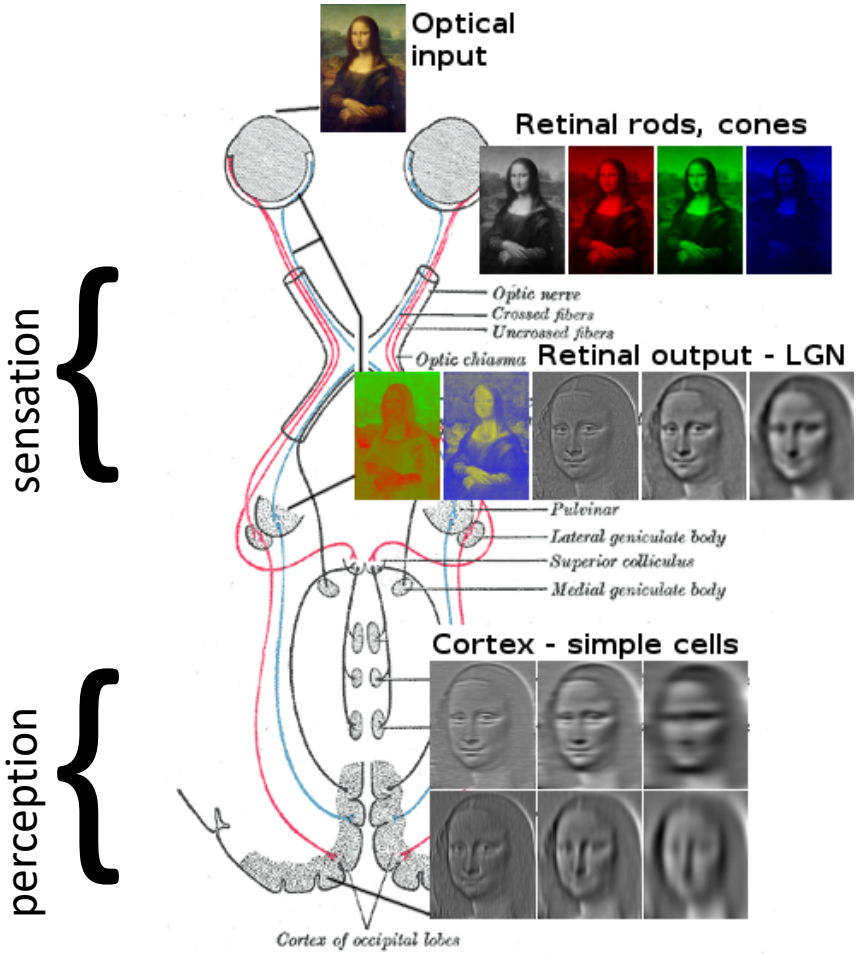
Like comics, visualizations let us abstract away less-relevant details...



The eye and the visual process

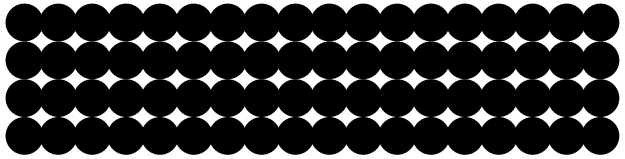


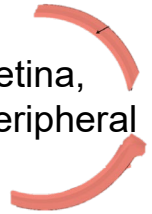




Cones and rods

Rods





Retina,
Peripheral

120 milions

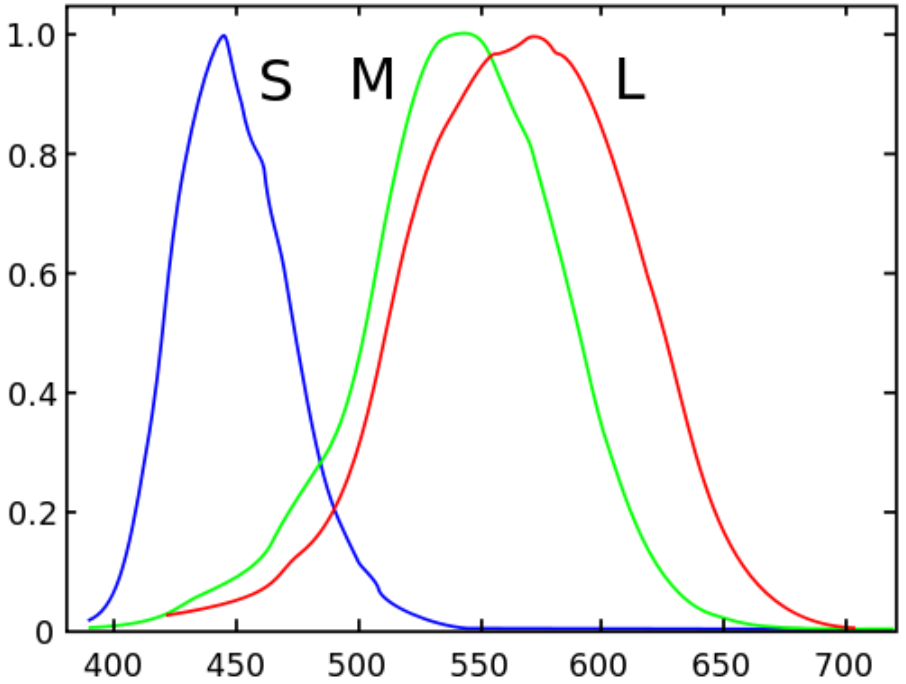
Cones





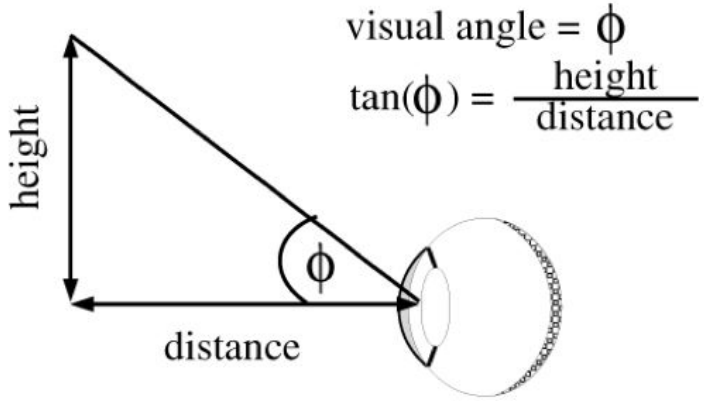
Fovea,
Central

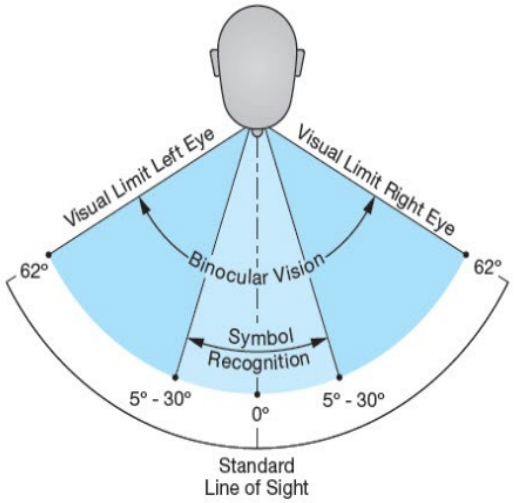
6.4 milions

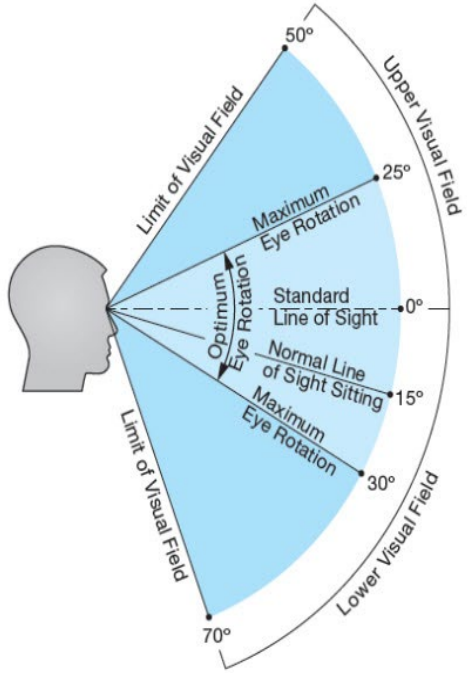


Visual angle and Useful field of view

Visual angle

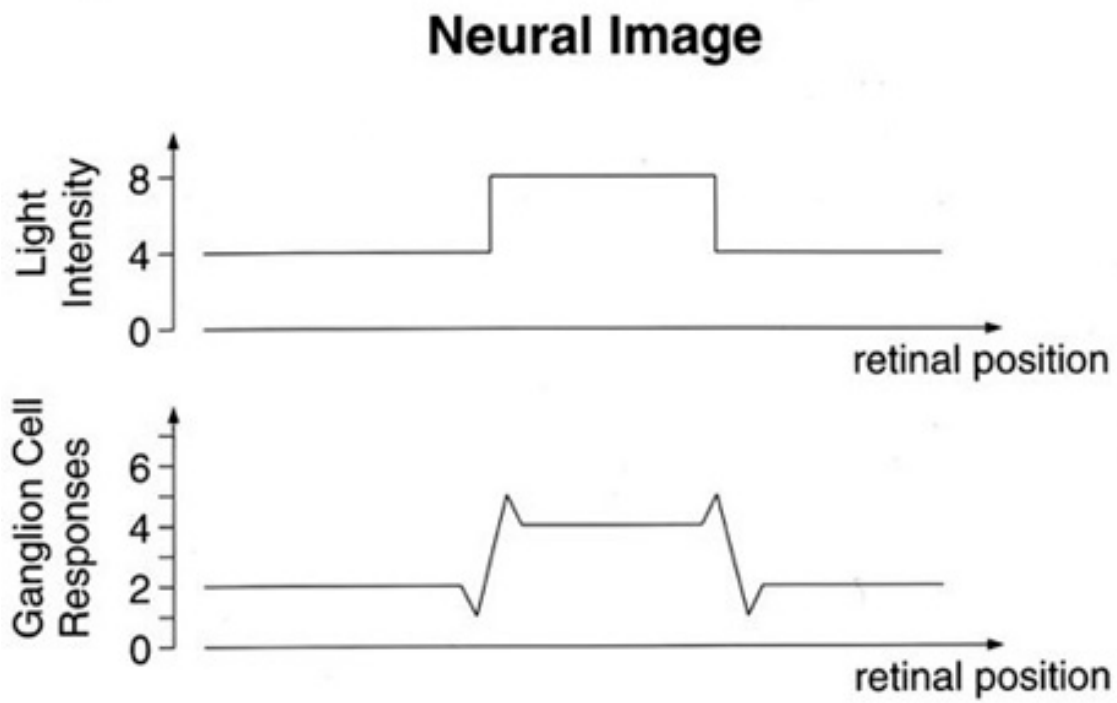


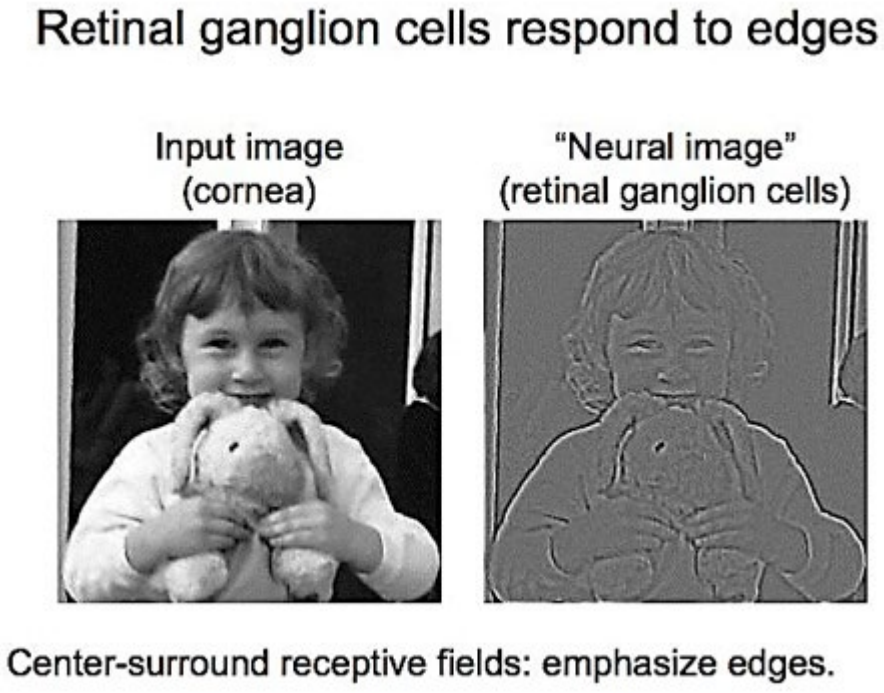




Source: http://www.cns.nyu.edu/~david/courses/perception/lecturenotes/eye/eye.html

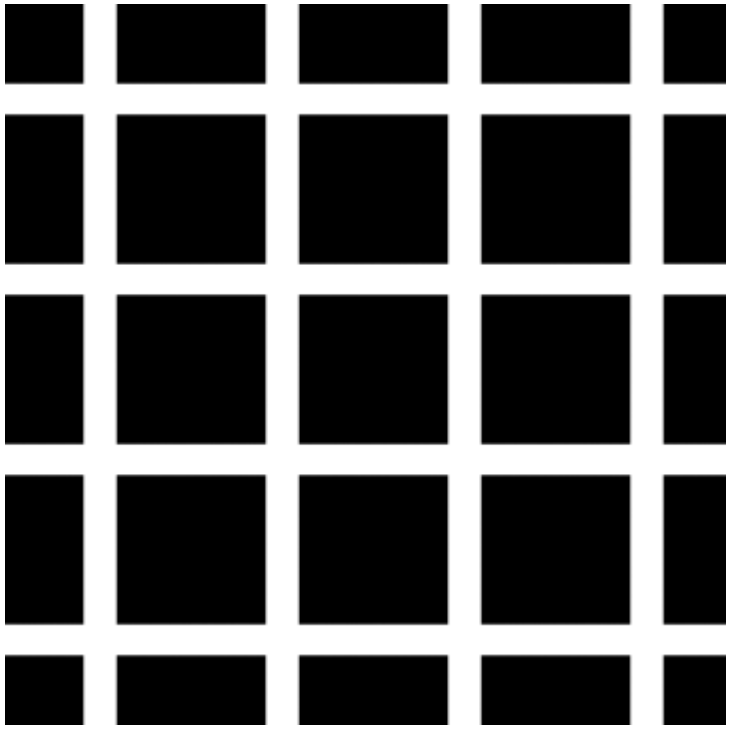
Economy of effort



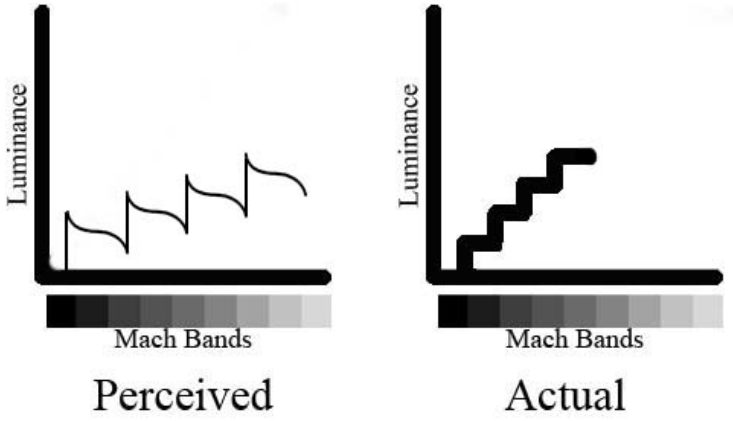


Fuente: Heeger, 2006

Biased signals sent to neurons

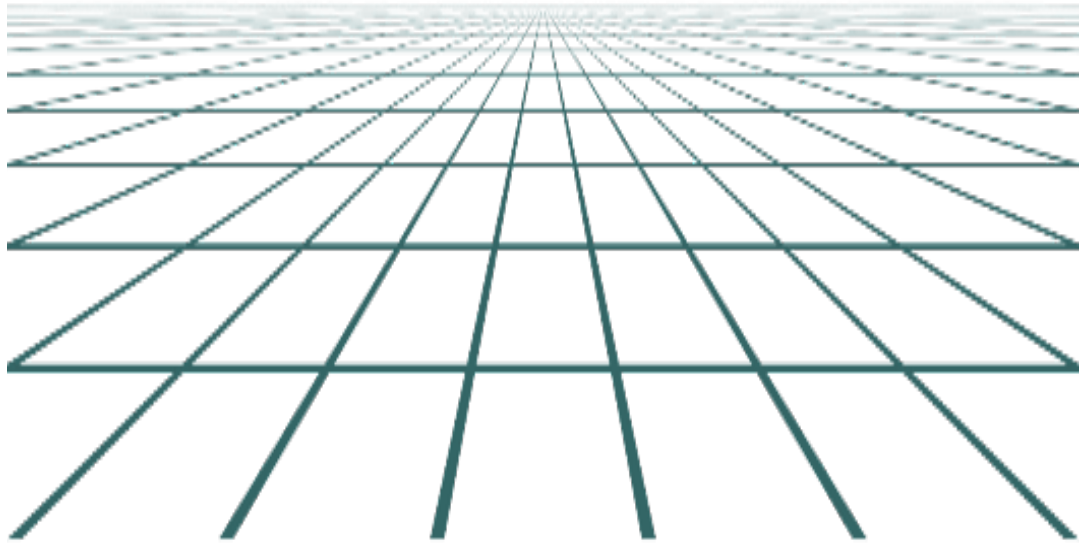


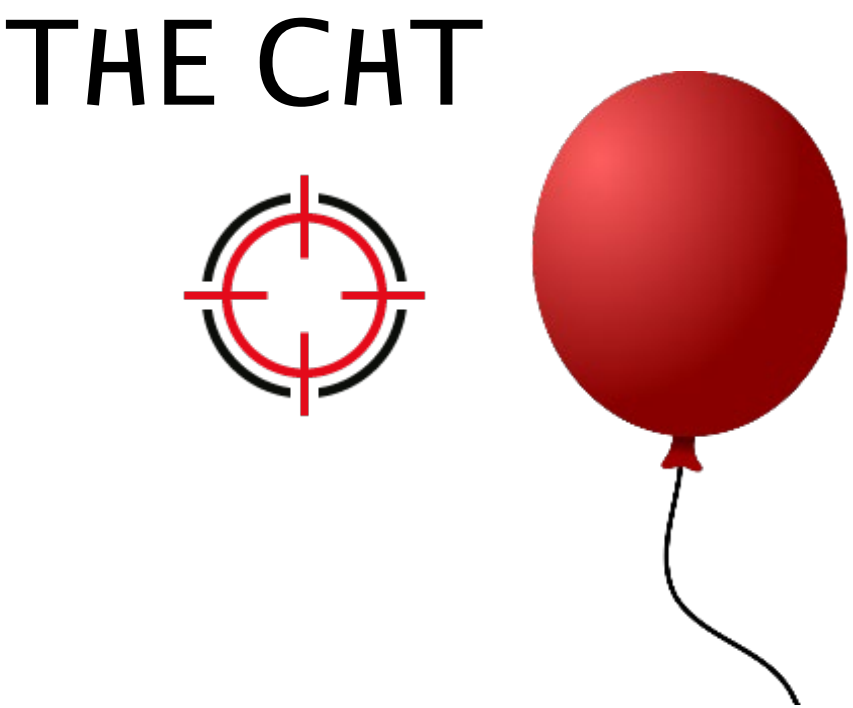




From eyes or from mind?

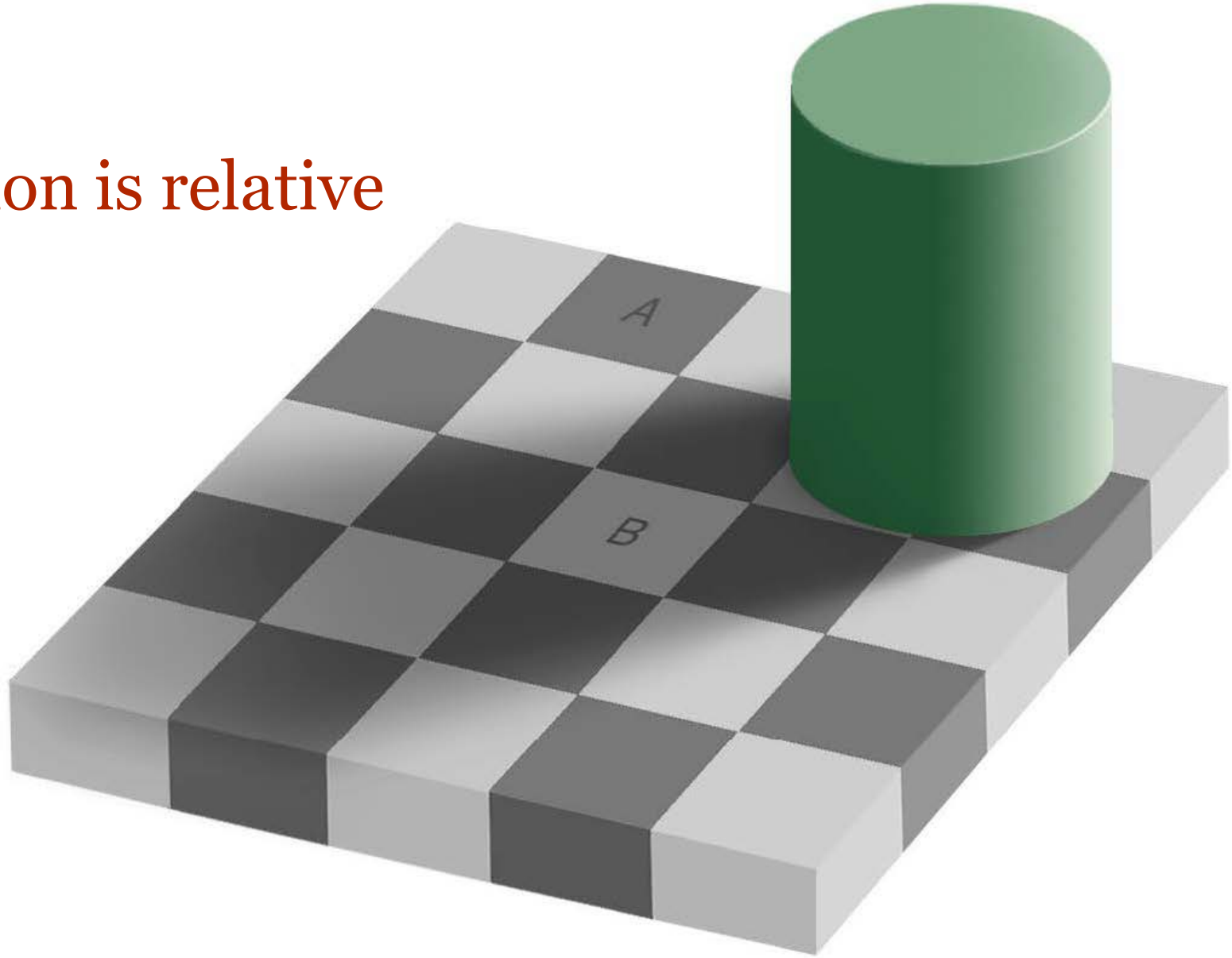
iStock.com/ ImpaKPro



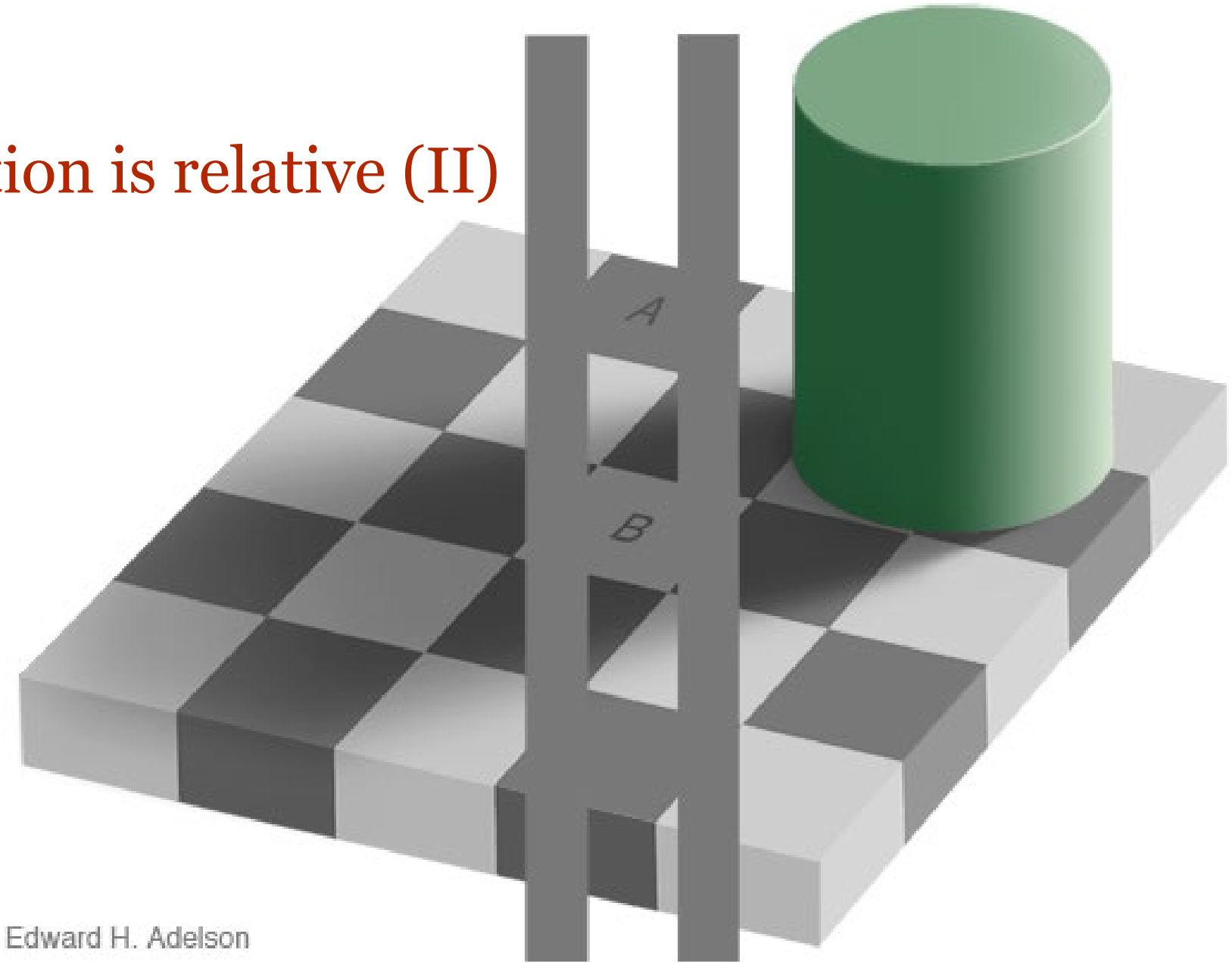


iStock.com/Sigit Mulyo Utomo Anastasiia_M

Perception is relative

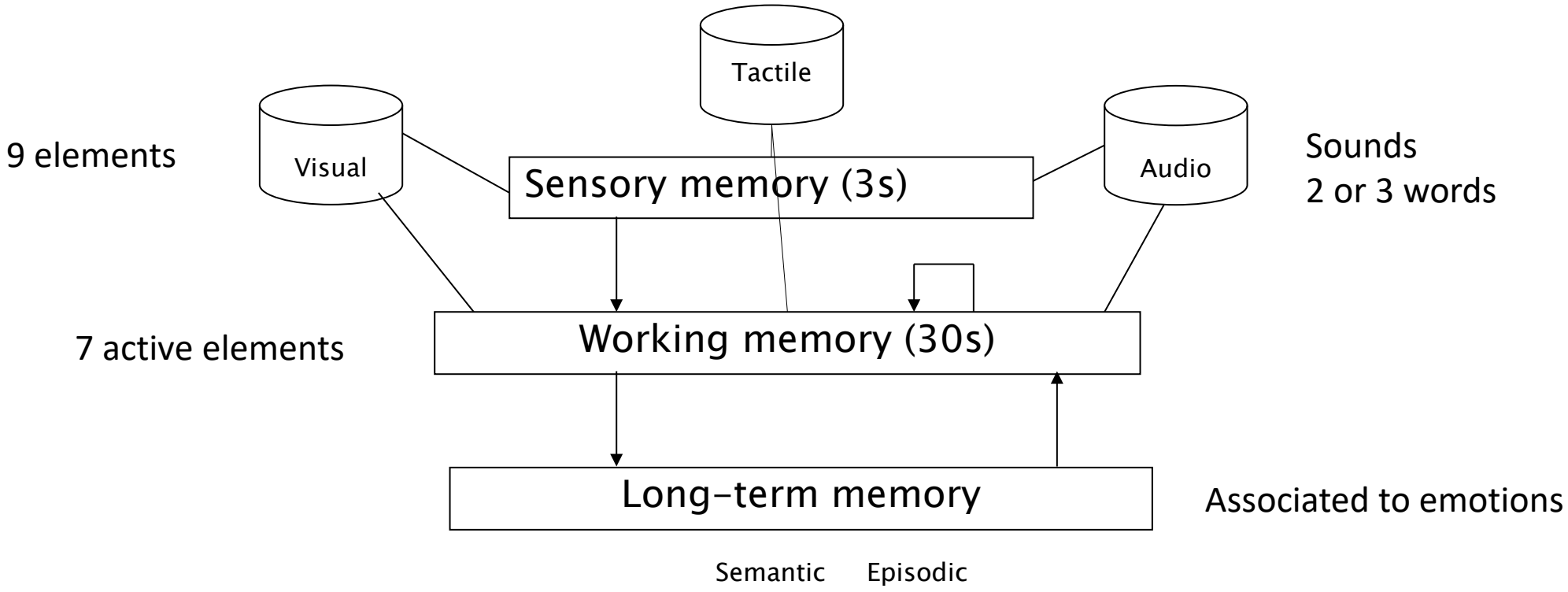


Perception is relative (II)



Edward H. Adelson

Memory


9 elements
Tactile
Visual
Audio
Sounds
2 or 3 words
Sensory memory (3s)
7 active elements
Working memory (30s)
Long-term memory
Associated to emotions
Semantic
Episodic

Try to remember

724103582910643512 90

498 745 970 231 443 671 88

What do you have to take first on an ATM: money or card? Why?

Attention

- Focused attention
  - All our attention in one event
- Divided attention
  - Our attention shifts between two or more events
  - Be careful with balance and interferences

We want to optimize...

Differentiation

Magnitude estimation

Signal detection (preattentive)

Selective attention

Change blindness

Cognitive effort (Gestalt)

We want to optimize...

Differentiation

Discriminality: how easy is to tell two things apart?

Magnitude estimation

Signal detection (preattentive)

Selective attention

Change blindness

Cognitive effort (Gestalt)

Which is brighter?



Rgb(128,128,128)

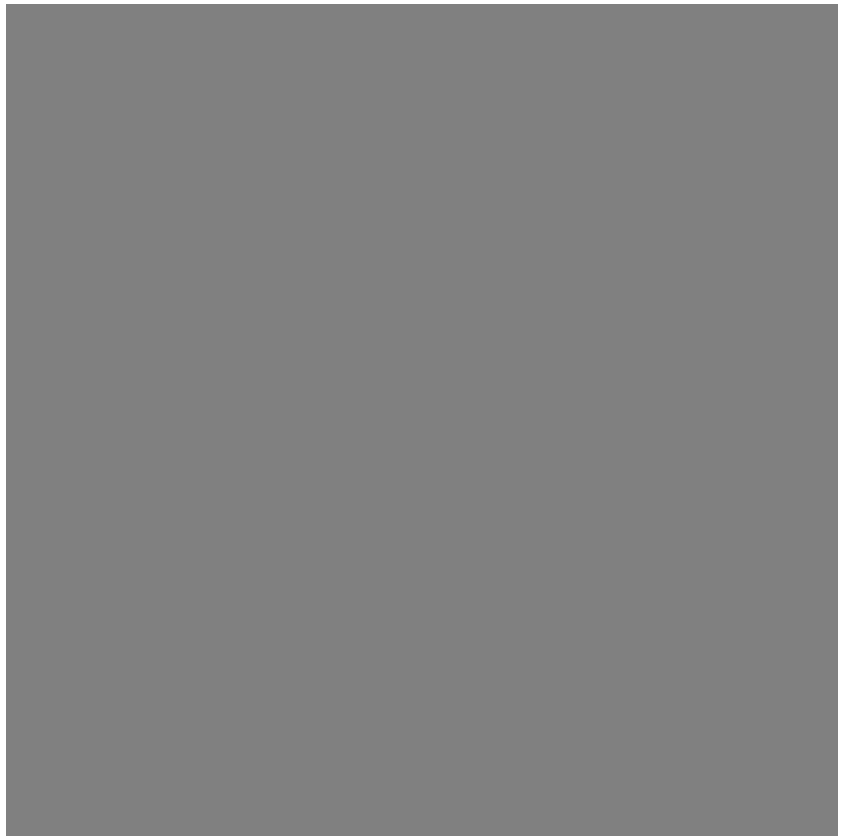


Rgb(144,144,144)

Which is brighter?



Rgb(134,134,134)



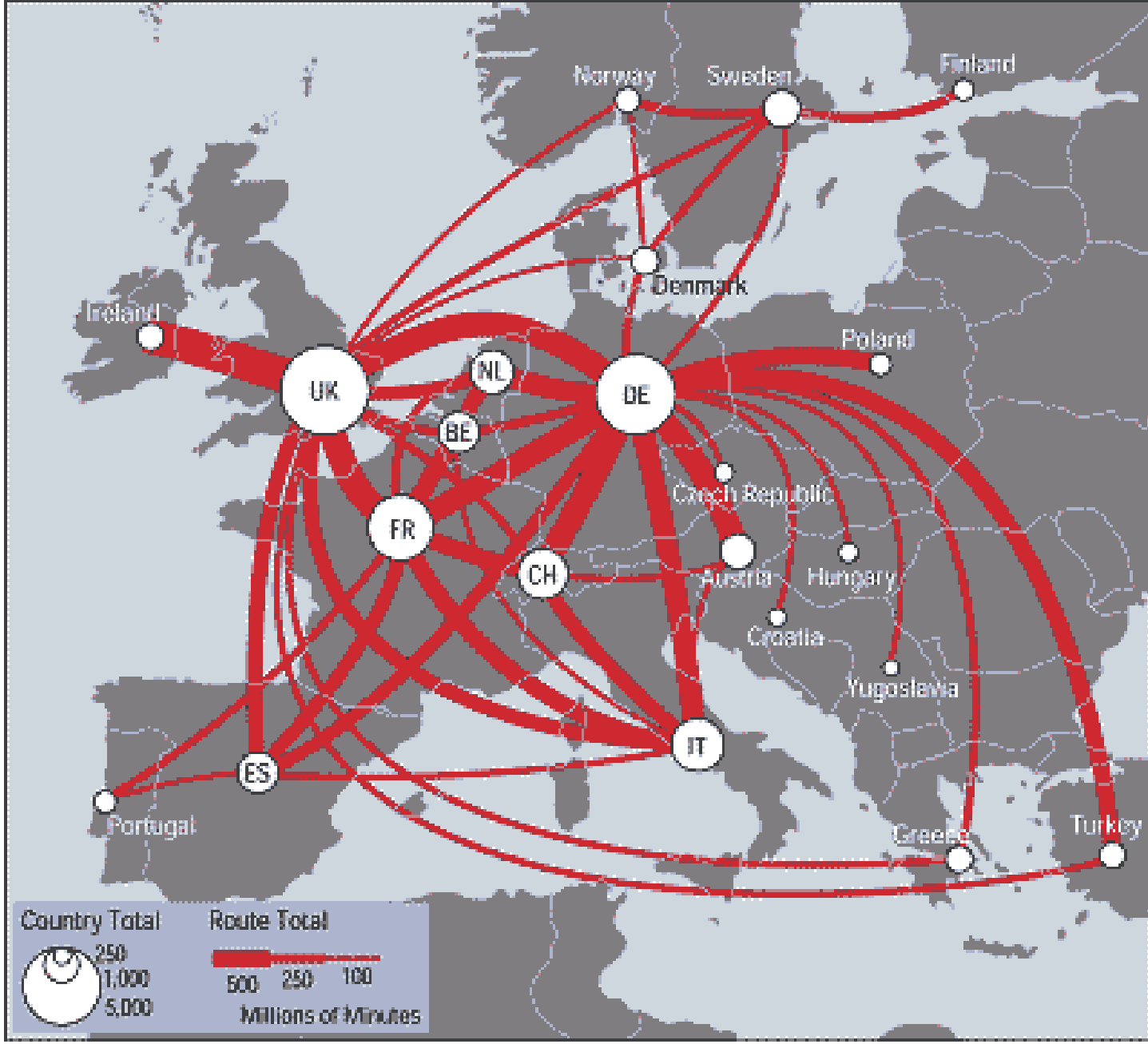
Rgb(128,128,128)

RULE 1. Better discrete scales than continuous

Law of just noticeable difference, by Ernst Weber






Norway
Sweden
Finland
Denmark
Ireland
Poland
UK
NL
DE
BE
Czech Republic
FR
CH
Austria
Hungary
Croatia
Yugoslavia
IT
ES
Portugal
Greece
Turkey
Country Total
250
1,000
5,000
Route Total
500
250
100
Millions of Minutes

We want to optimize...

Differentiation

Magnitude estimation

Signal detection (preattentive)

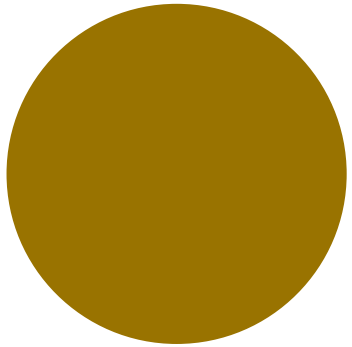
Selective attention

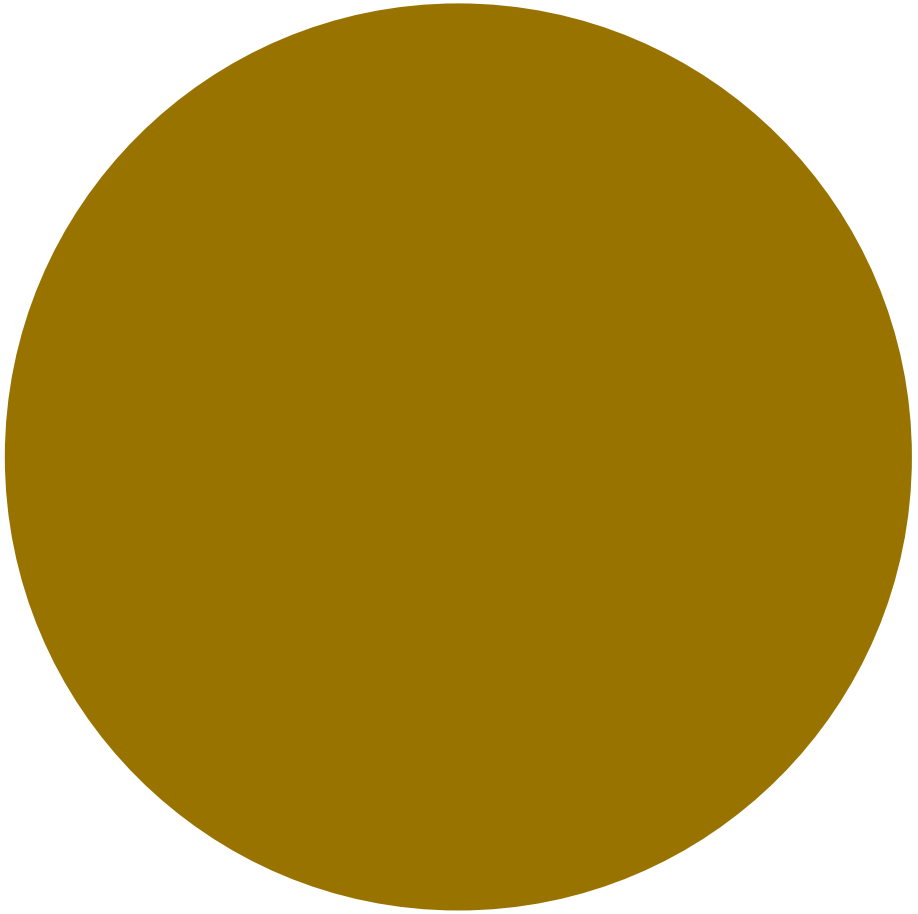
Change blindness

Cognitive effort (Gestalt)

Accuracy: How correctly can we read off values?

How much larger is the area of the big circle?

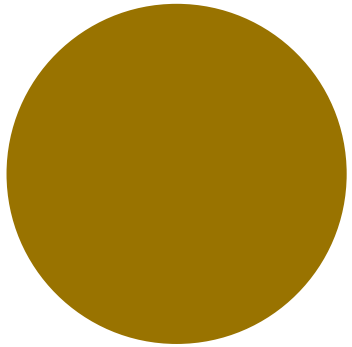


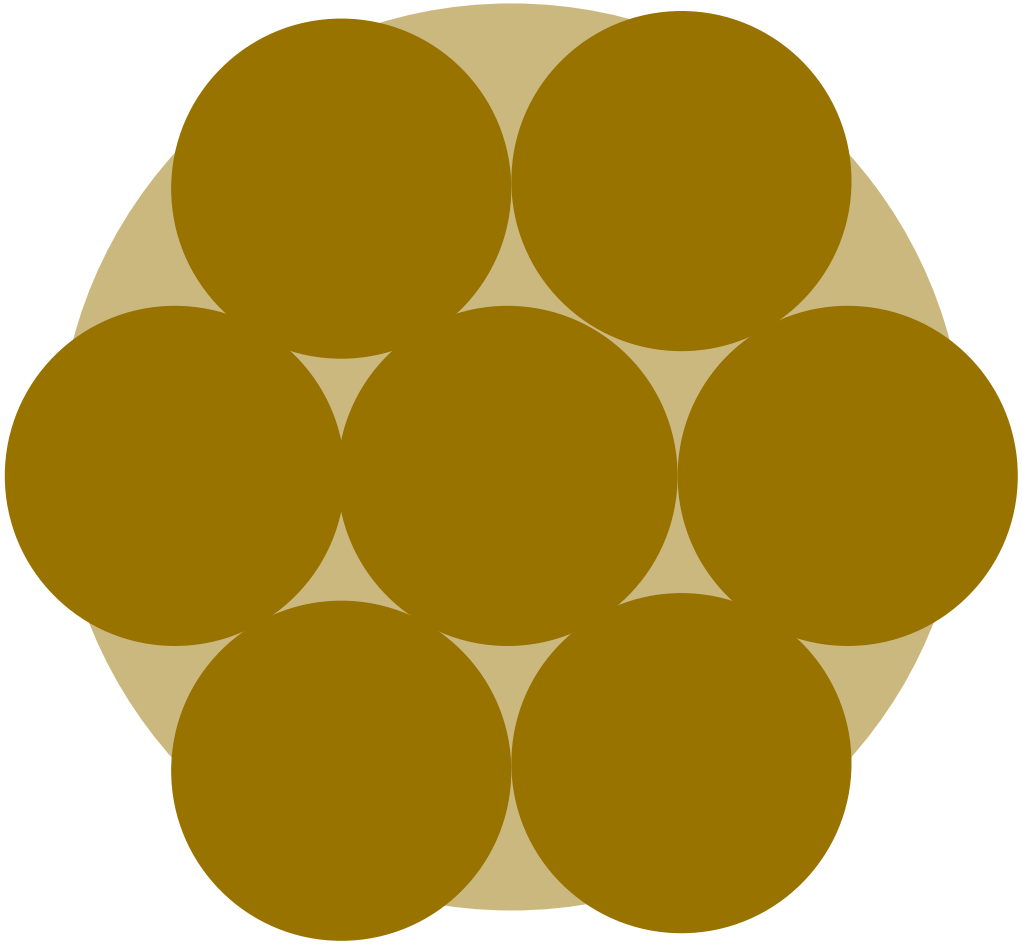


How much larger is the big bar?

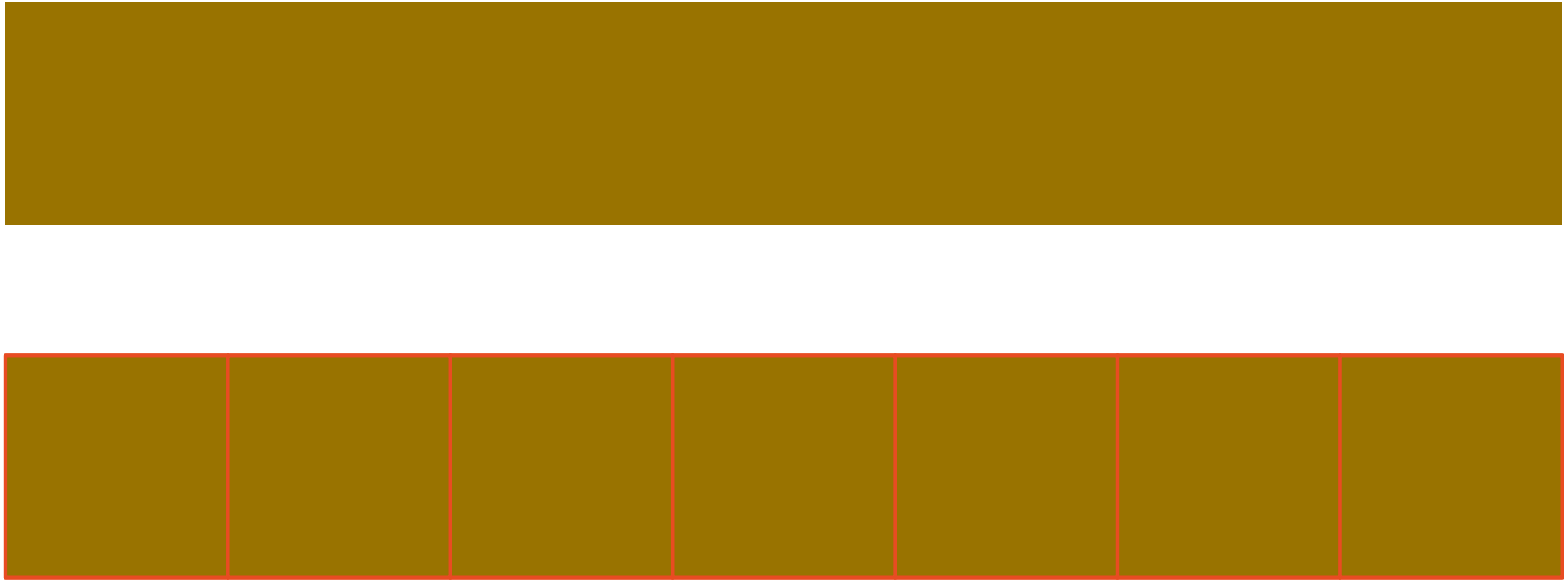


How much larger is the area of the big circle?

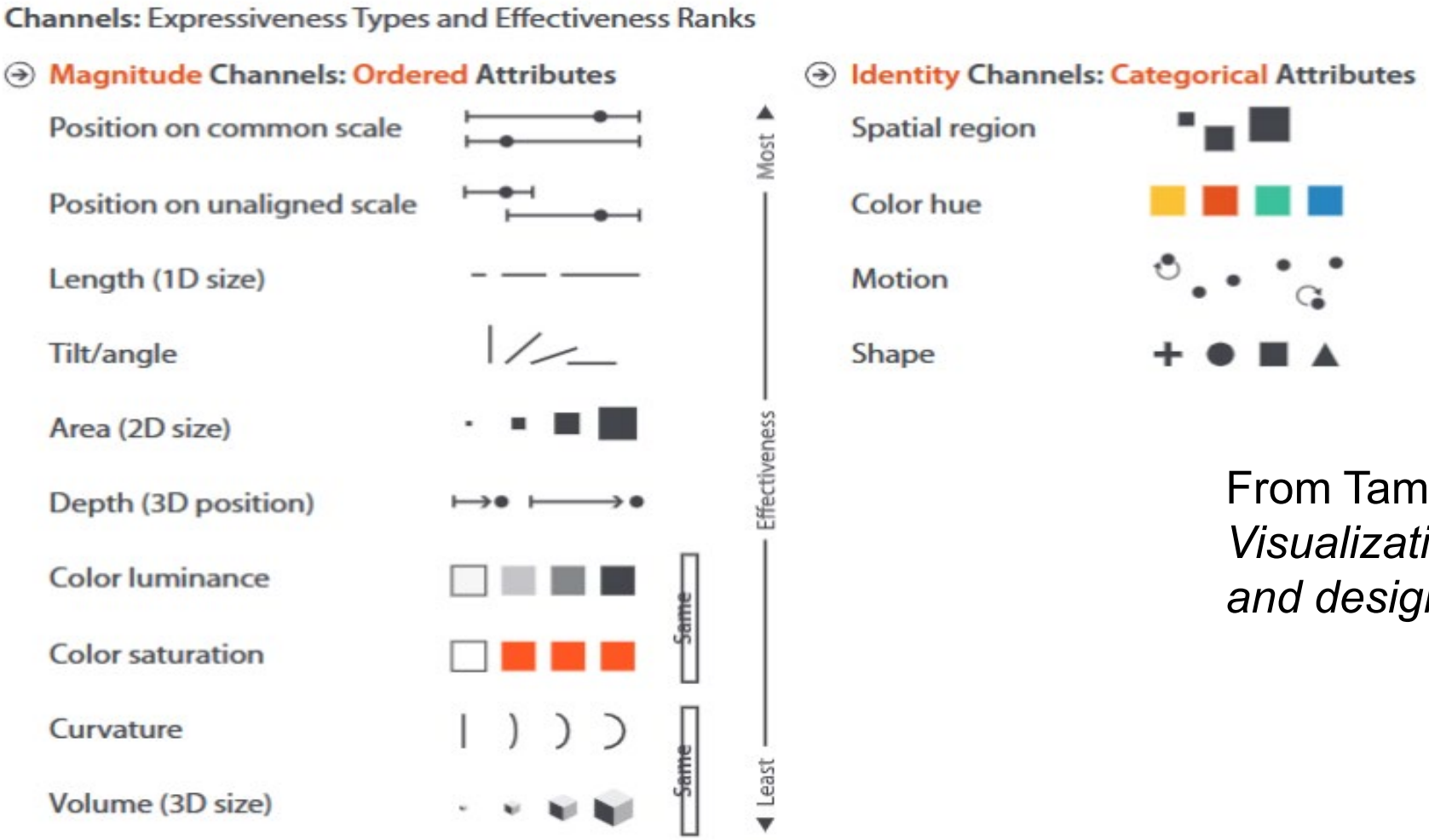




How much larger is the big bar?



RULE 2. Take effectiveness into account



From Tamara Munzner.
Visualization analysis and design

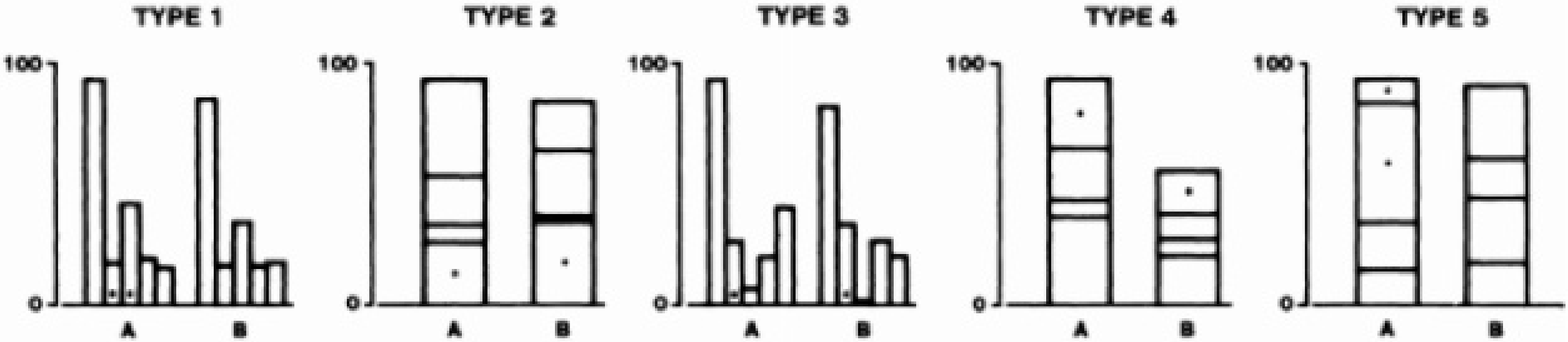
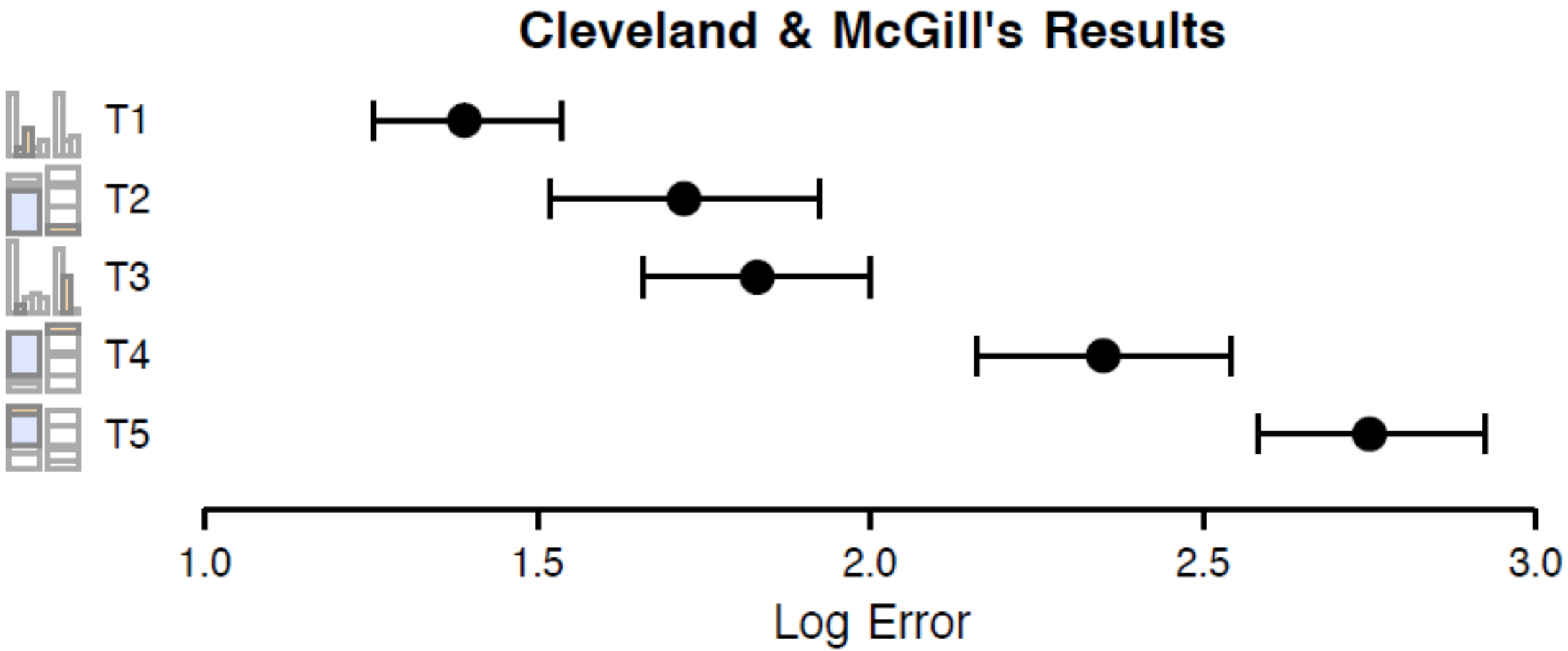
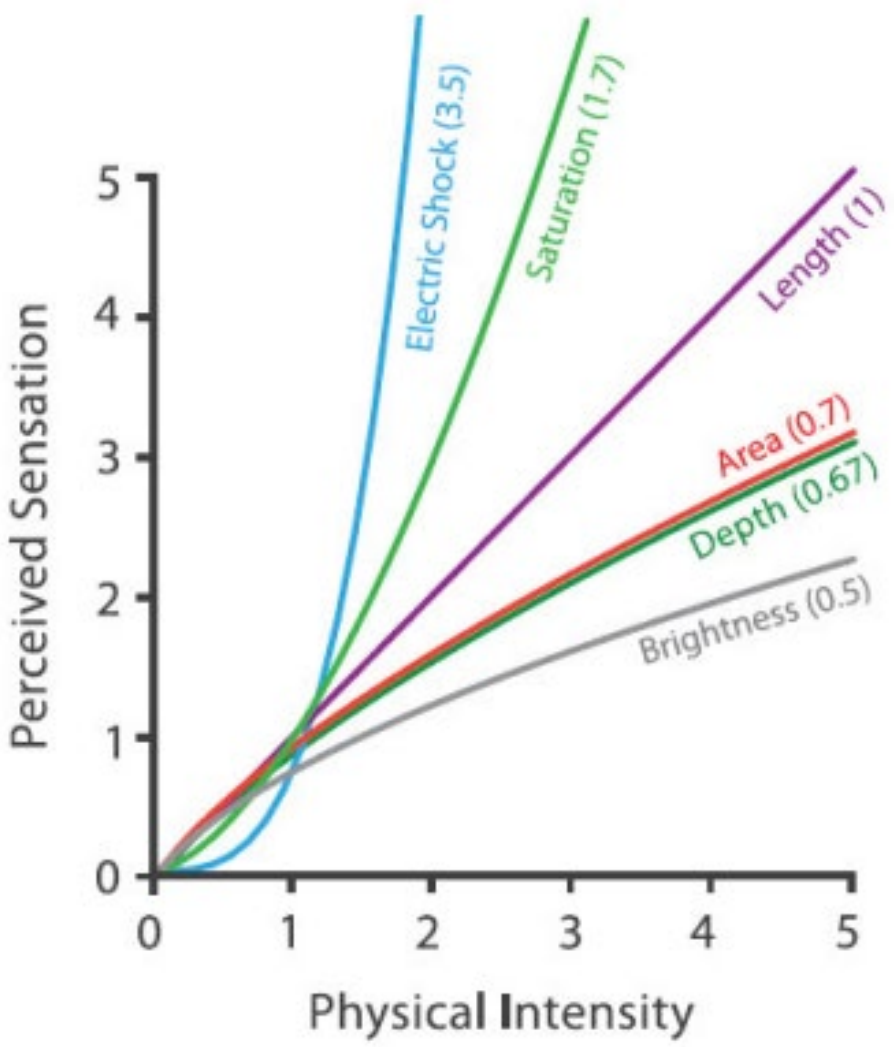


Figure 4. Graphs from position–length experiment.

Cleveland W and McGill R. Graphical Perception: Theory, experimentation and applications to the development of graphical methods. JASA 79 (387): 531-554; 1984.


Cleveland & McGill's Results
T1
T2
T3
T4
T5
1.0
1.5
2.0
2.5
3.0
Log Error

Innate biases in perception of change



From visualization analysis and design, by tamara Munzner

We want to optimize...

Differentiation

Magnitude estimation

Signal detection (preattentive)

Selective attention

Change blindness

Cognitive effort (Gestalt)

Pop out: how easy is to spot some values from the rest?

How many 5 are there?

38572093982372819683729382 7

38291235838349273012289483 9

90902010203289375927309142 8

93830976296581743186924102 4

How many 5 are there? (II)

385720939823728196837293827

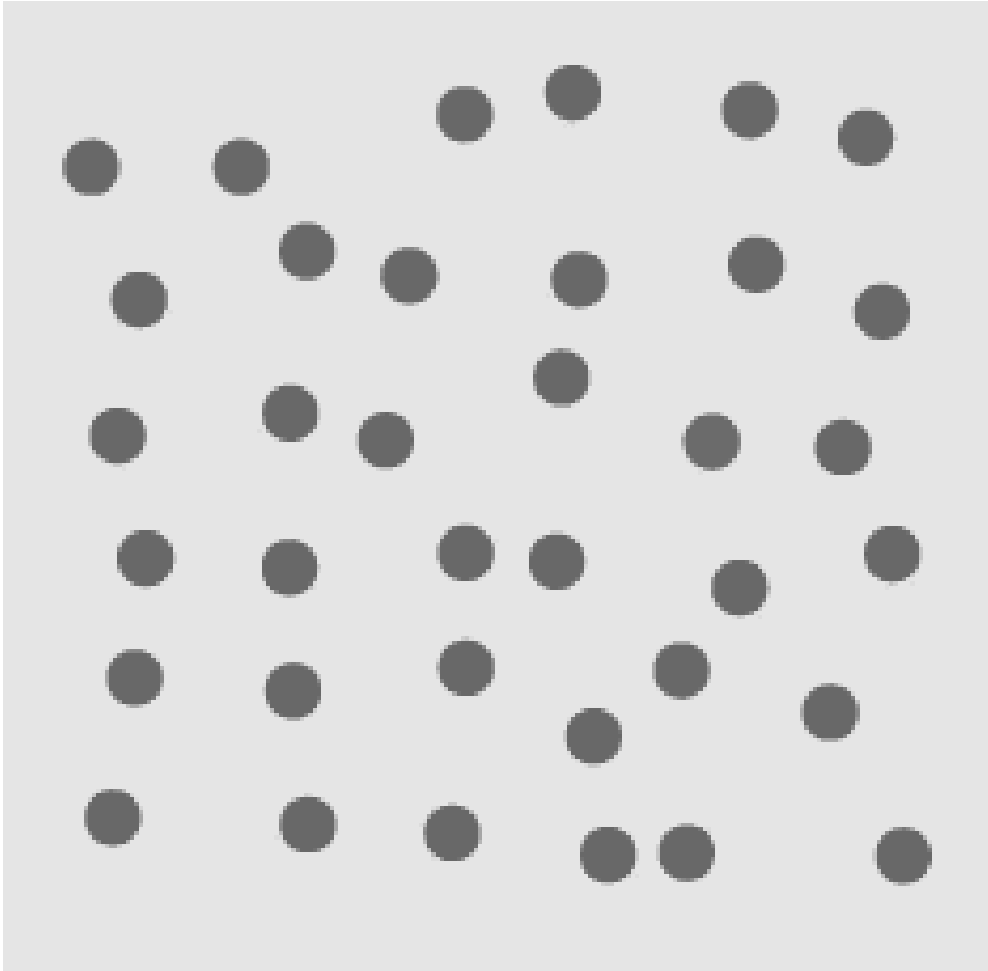
382912358383492730122894839

909020102032893759273091428

938309762965817431869241024

Preattentive properties

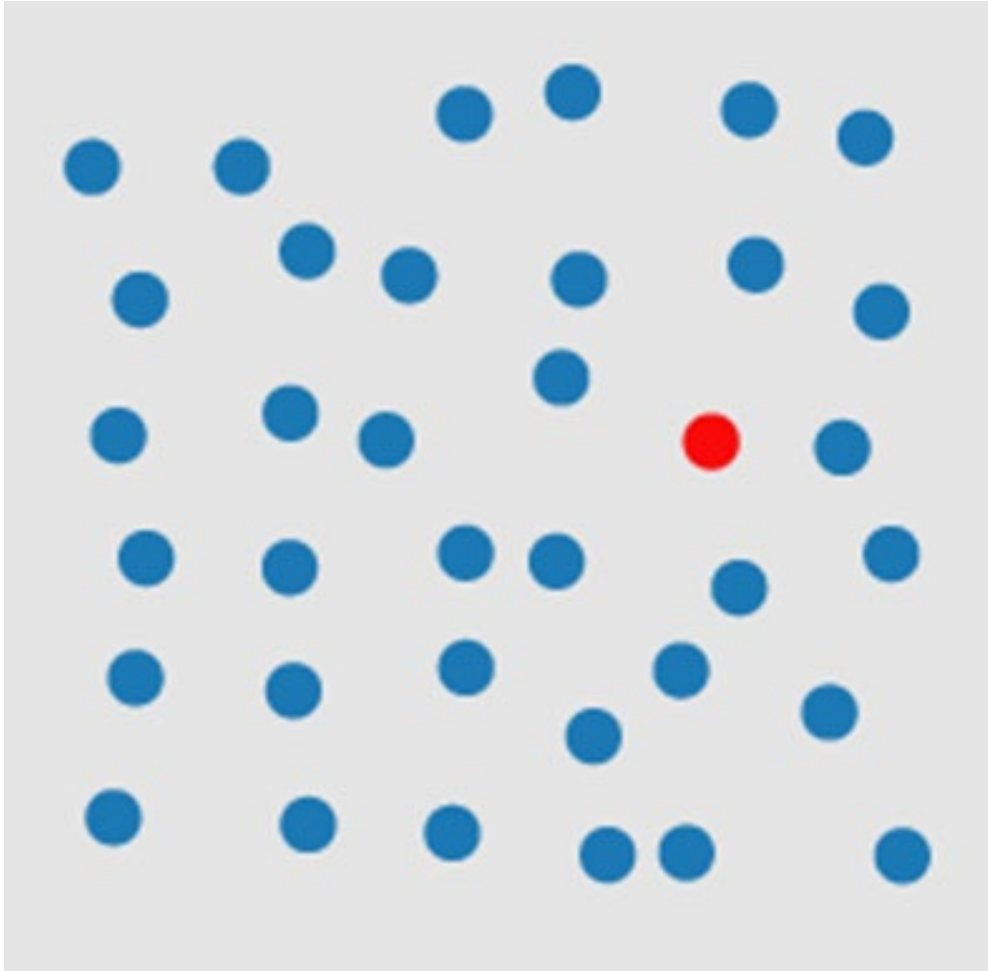
- Immediately detected
- No serial search
- No distractors



Source: Healey, 2012

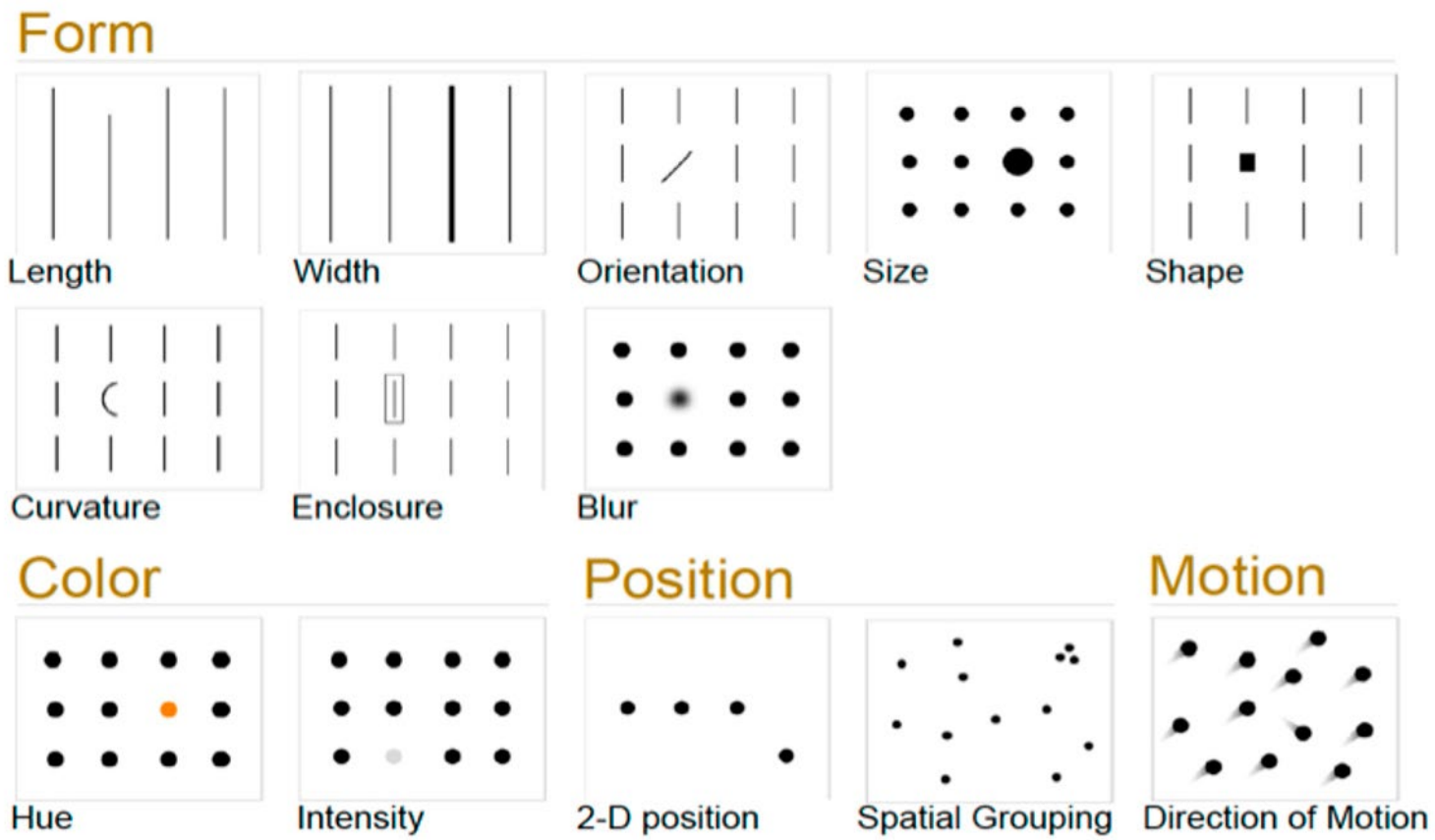
Source: C. Healey and J. Enns, "Attention and Visual Memory in Visualization and Computer Graphics," in *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 7, pp. 1170-1188, July 2012, doi: 10.1109/TVCG.2011.127

Where is the red circle?



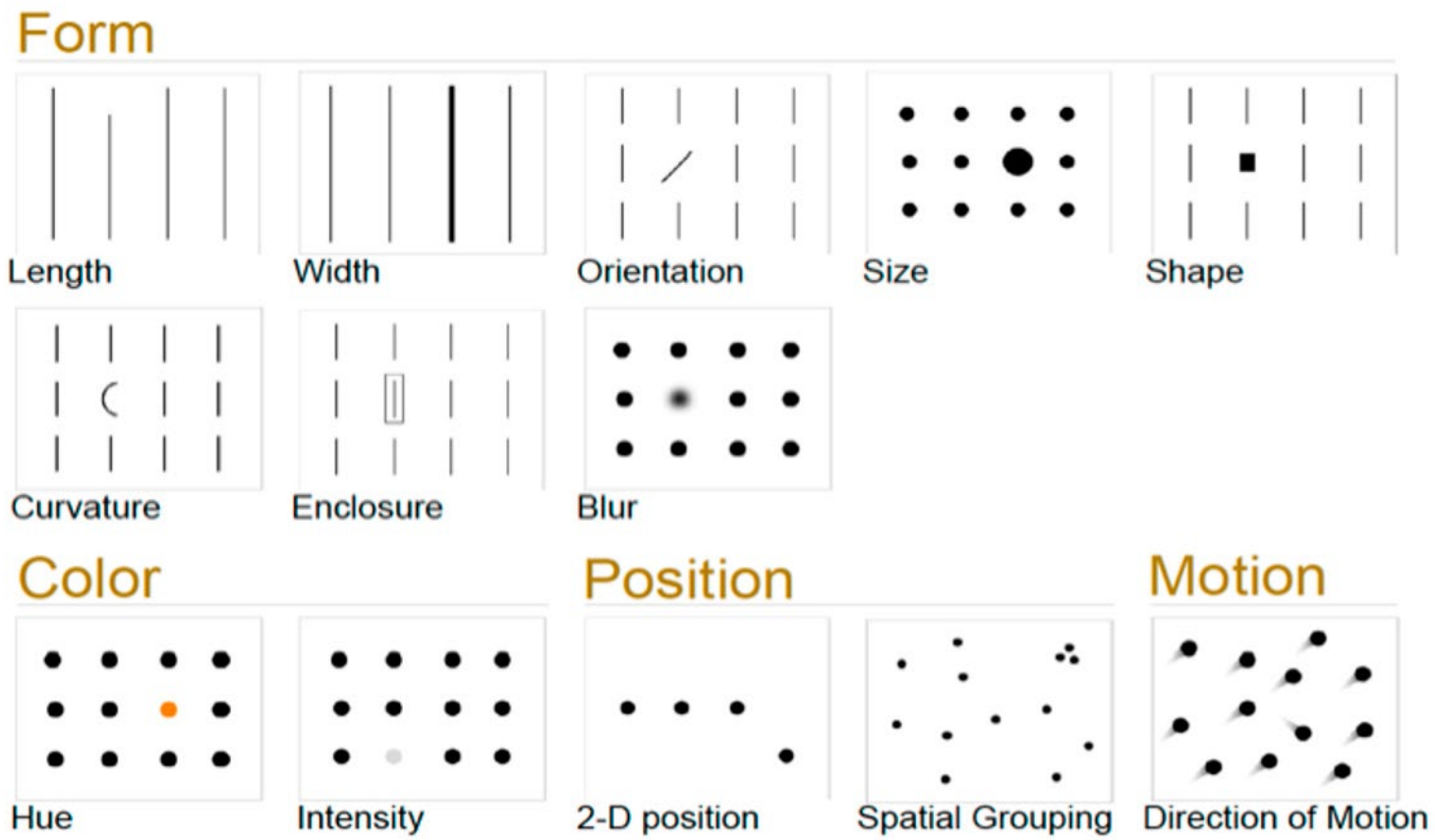
Source: Healey, 2012

Preattentive properties. Types



Source: Few, 2012

Preattentive properties. Types



Source: Few, 2012

We want to optimize...

Differentiation

Magnitude estimation

Signal detection (preattentive)

Selective attention

Change blindness

Cognitive effort (Gestalt)

Separability: how much interaction occurs between attributes

One dimension: lightness







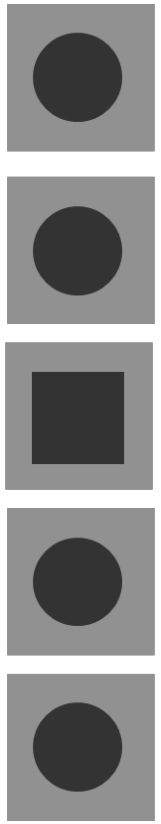


One dimension: shape









Redundant: shape and lightness

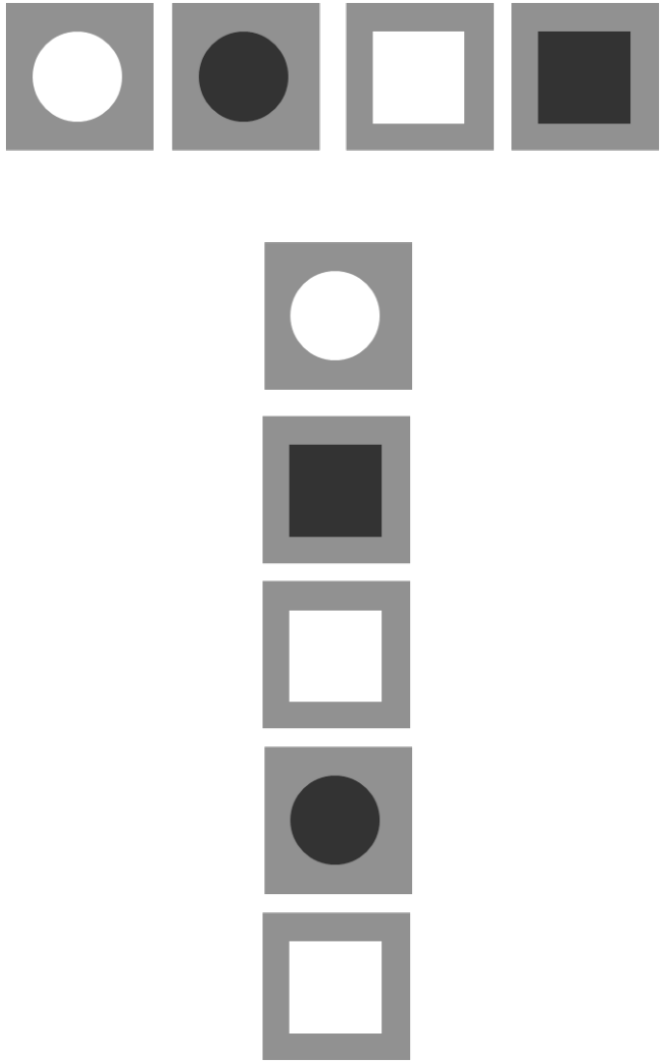




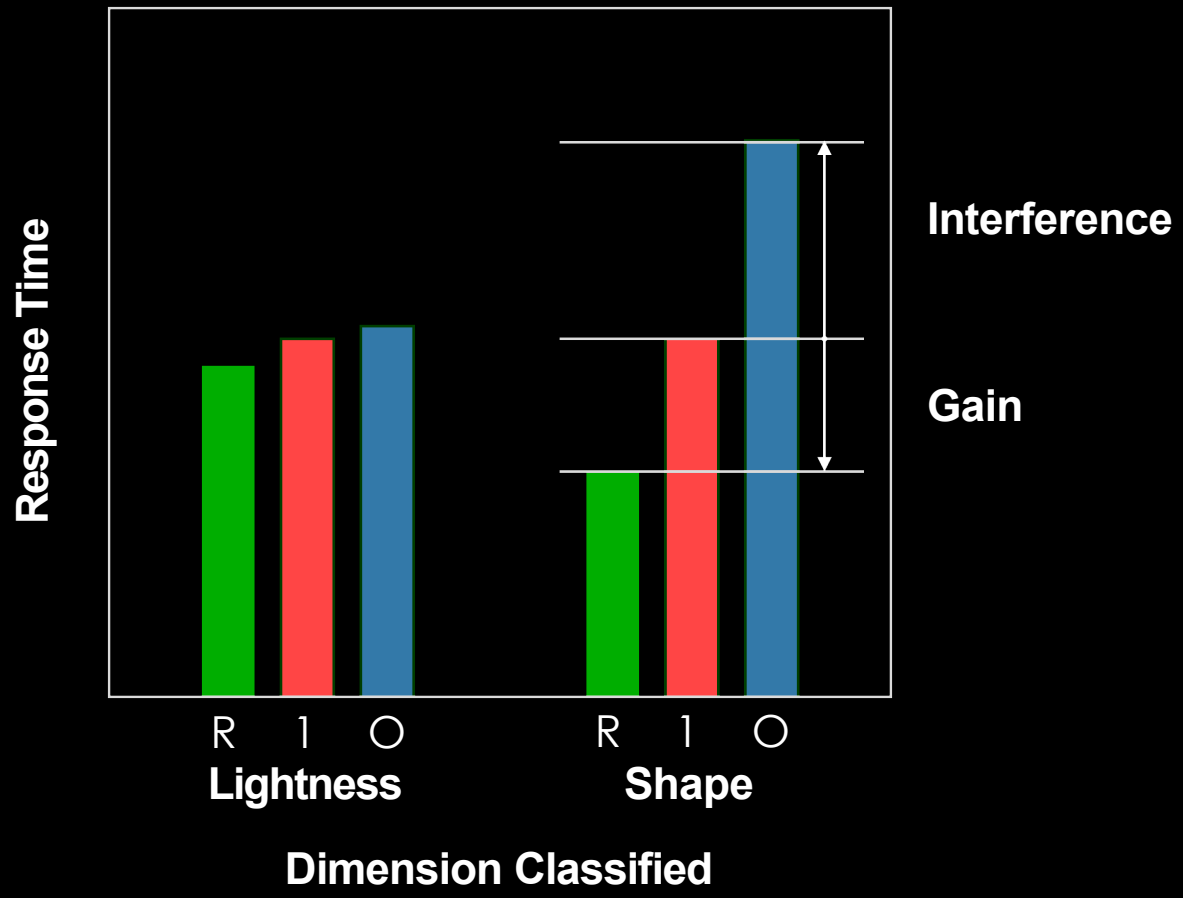




Orthogonal: shape and lightness



Orthogonal: shape and lightness



We want to optimize...

Differentiation

Magnitude estimation

Signal detection (preattentive)

Selective attention

Change blindness

Cognitive effort (Gestalt)

Attention: Do we process all the information?




STATES
OF AMERICA
20
A 97567556 D
20
20
20

We want to optimize...

Differentiation

Magnitude estimation

Signal detection (preattentive)

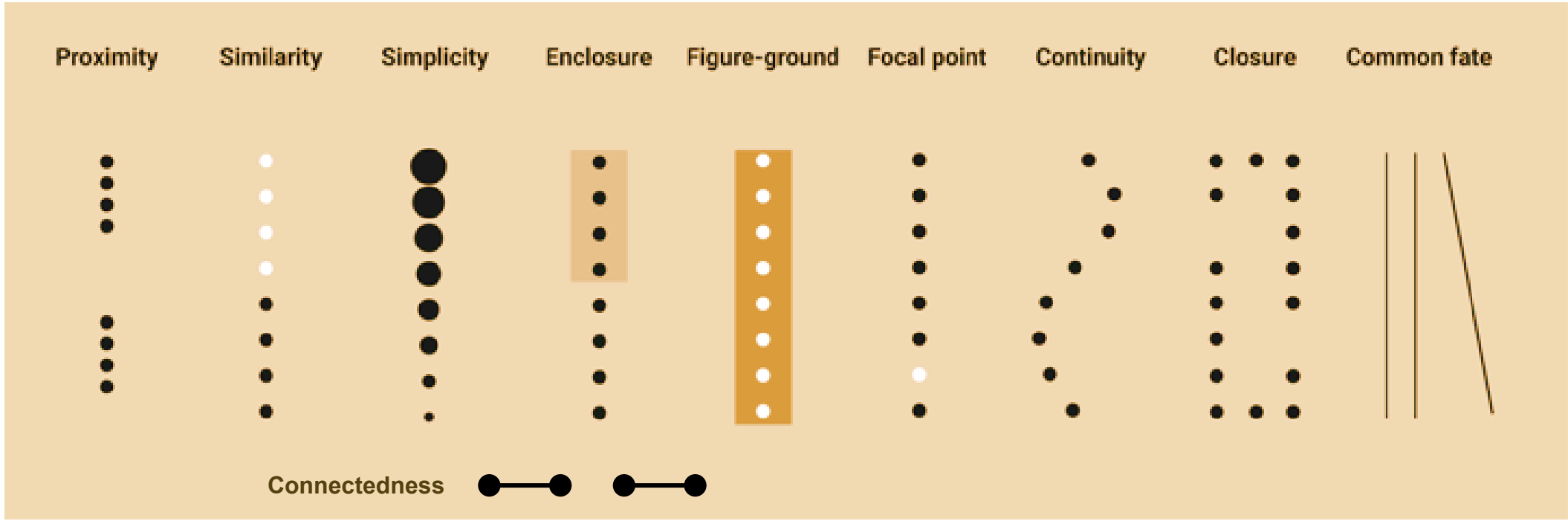
Selective attention

Change blindness

Cognitive effort (Gestalt)

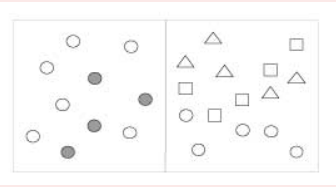
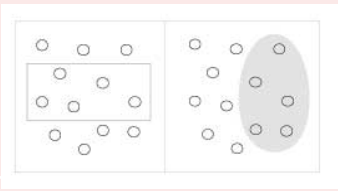
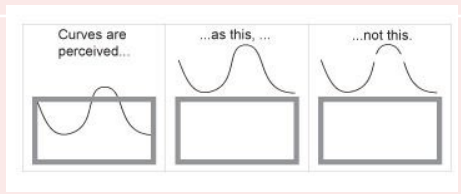
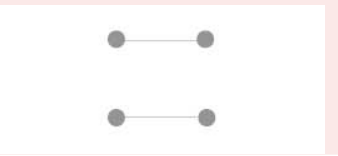
Interpretation: we favour the simplest

Gestalt laws



Source: van Dijk, 2022

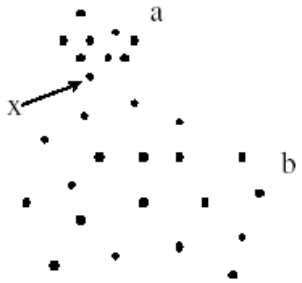
Gestalt laws

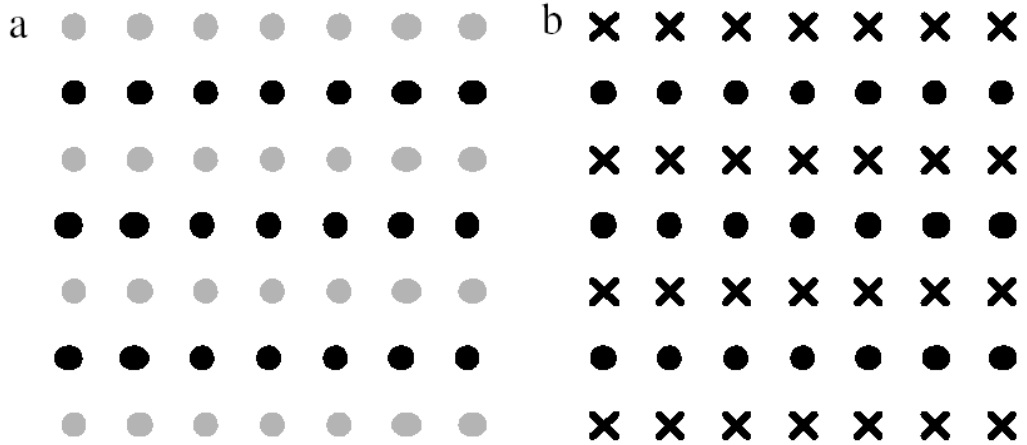
Proximity	Objects that are close together are perceived as a group.	
Similarity	Objects that share similar attributes (e.g., color or shape) are perceived as a group.	
Enclosure	Objects that appear to have a boundary around them (e.g., formed by a line or area of common color) are perceived as a group.	
Closure	Open structures are perceived as closed, complete, and regular whenever there is a way that they can be reasonably interpreted as such.	
Continuity	Objects that are aligned together or appear to be a continuation of one another are perceived as a group.	
Connection	Objects that are connected (e.g., by a line) are perceived as a group.	

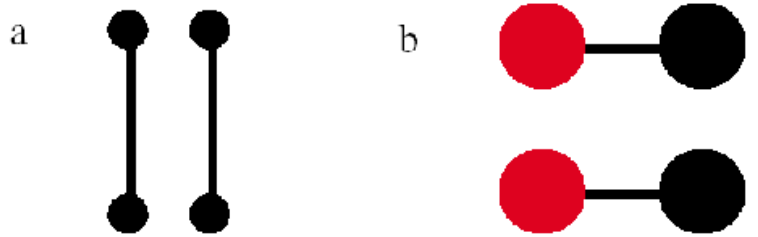
Source: Interaction Design Foundation
https://assets.interaction-design.org/literature/book/the-encyclopedia-of-human-computer-interaction-2nd-ed/data-visualization-for-human-perception

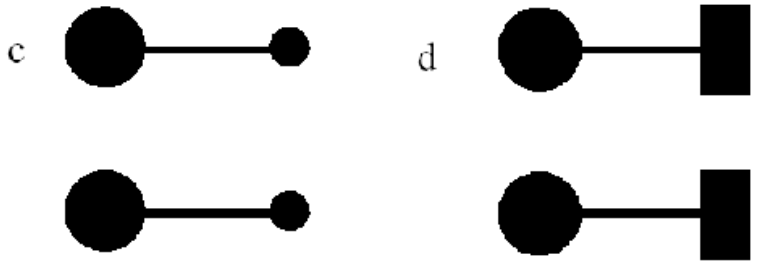
What do you see?

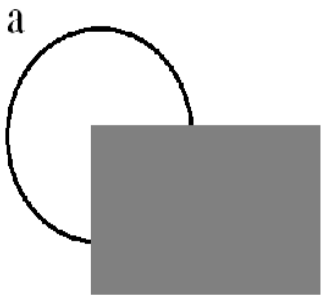


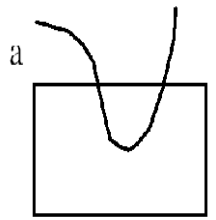




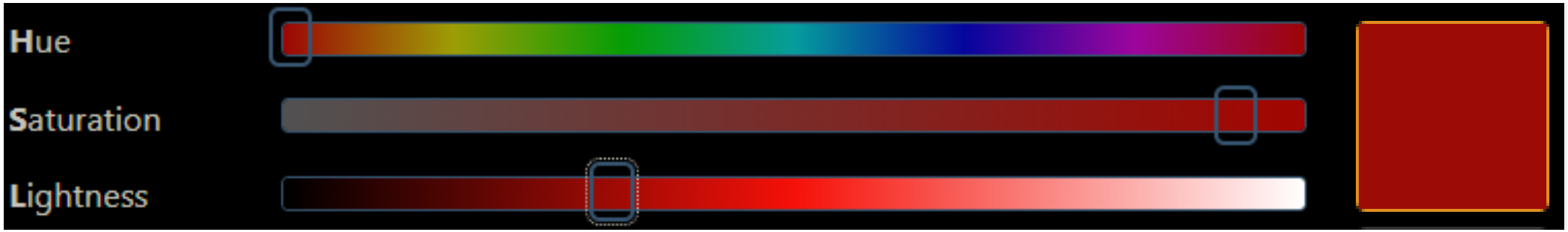








Colour dimensions



- **Hue o Chroma**, the name of the colour
- **Saturation**, the purity of colour
- **Luminosity**, whire/black level. Most accurately perceived

Colour blindness

Normal vision



Deuteranopia (green is not perceived)



Protanopia (red is not perceived)



Titranopia (blue is not perceived)



Achromatopsia (no colors perceived)



Source:Colour Blind Awareness

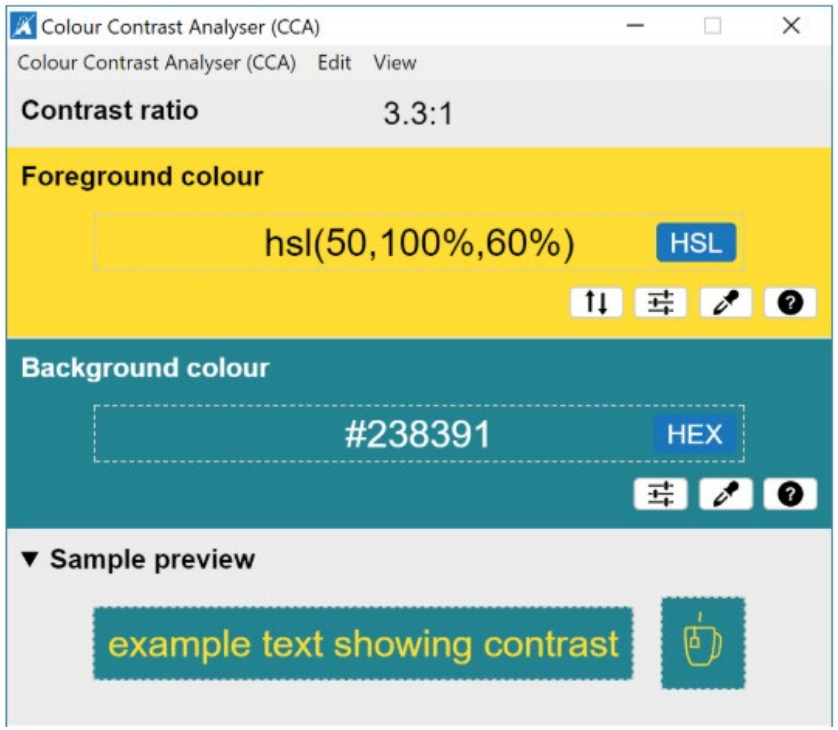
Age, Colour blindness and contrast

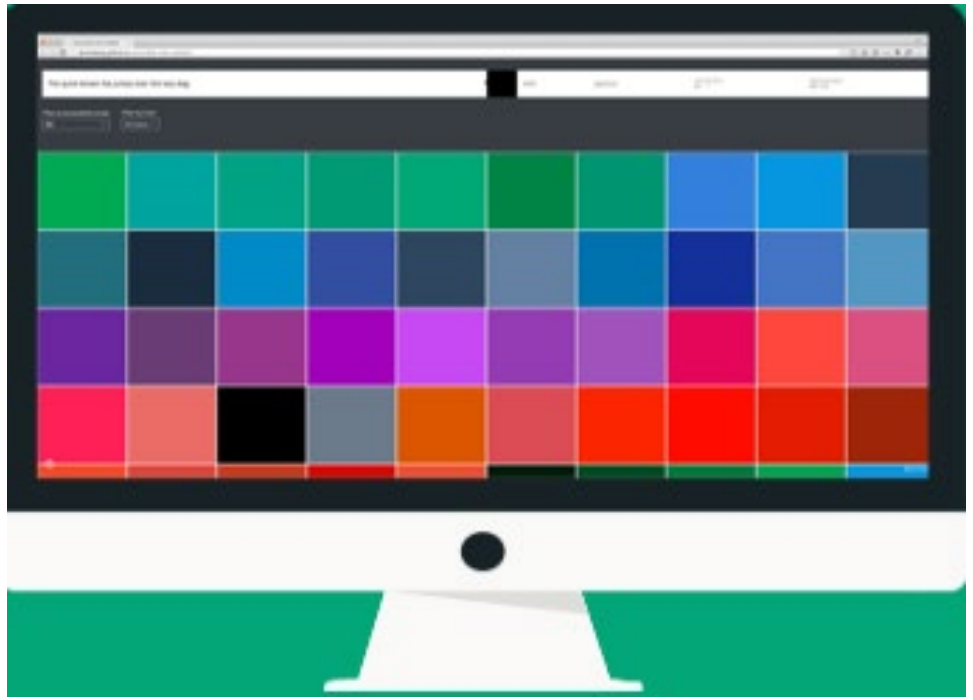
- Colour perception decreases with age
- Sight decrease affects both rods and cones
- As we have many more rods, elders perceive much better luminance differences than hue differences.
- It is therefore important to keep contrast differences in every colour system.

Class activity

- The Hue-test challenge http://www.xrite.com/hue-test/
- Simulating colour vision Silktide disability simulator

Contrast: some tools

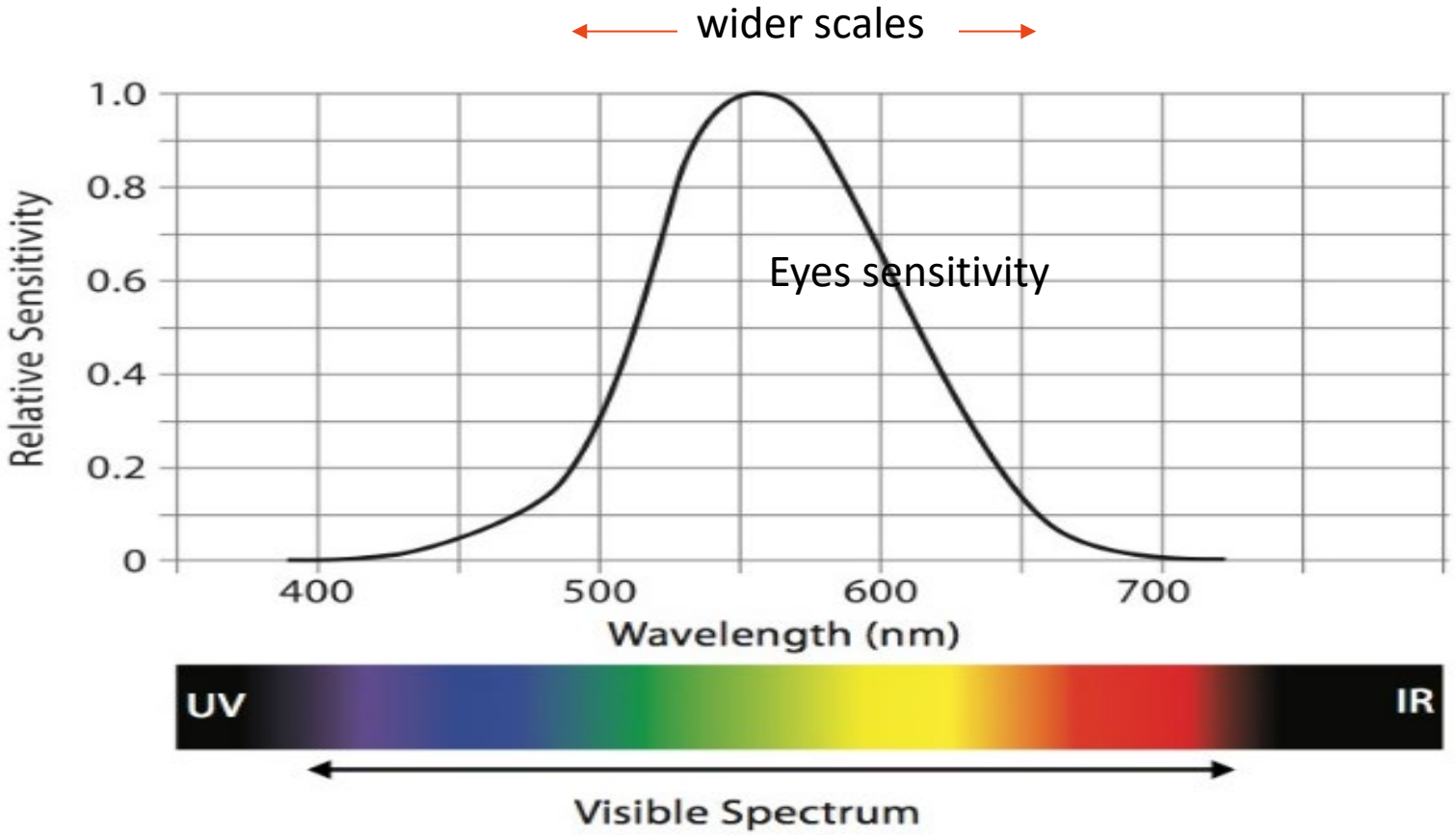


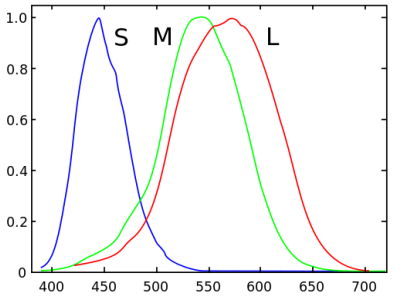


LINUX: https://contrast-ratio.com

Carbon IBM's Design System: Accessible colour palettes for information visualization

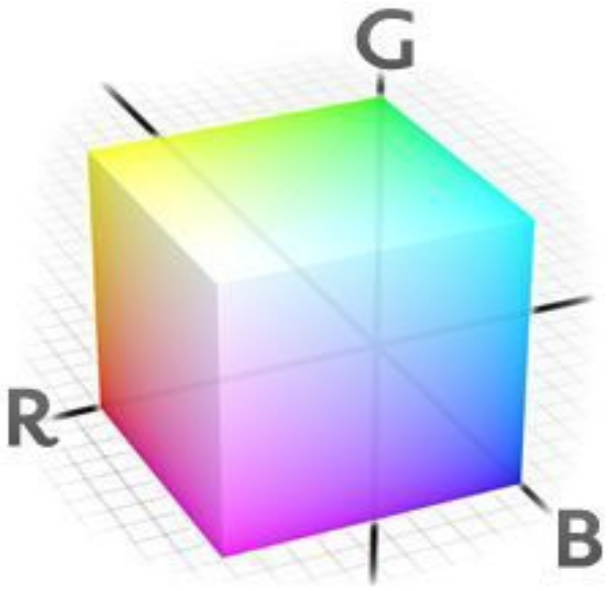
Eyes' sensitivity

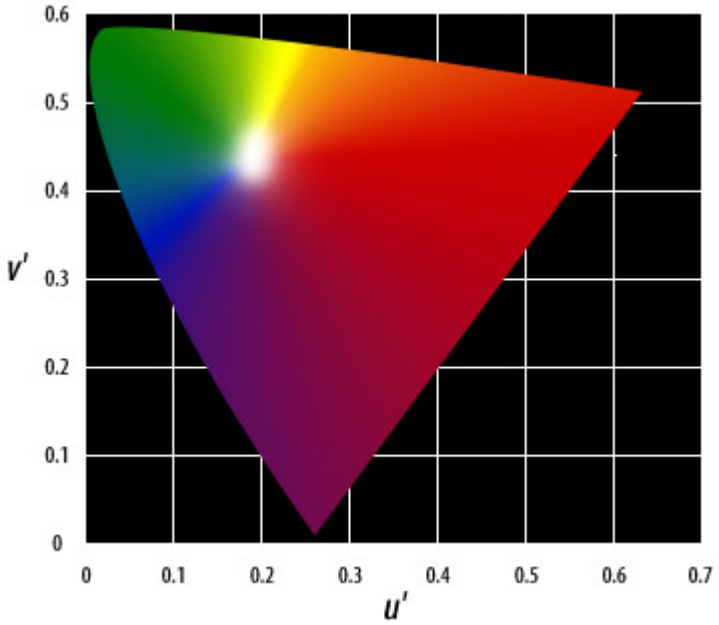


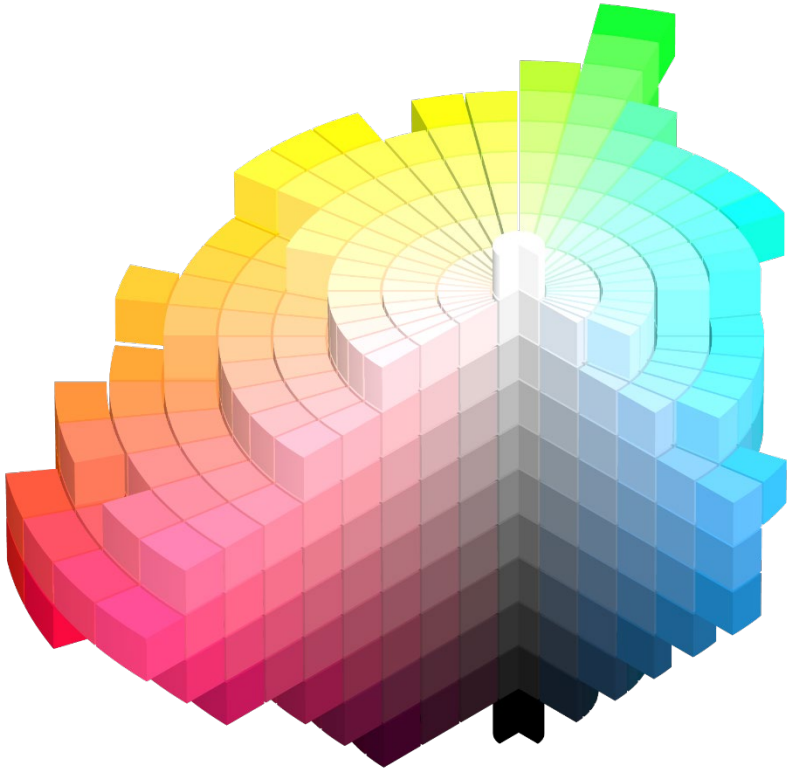




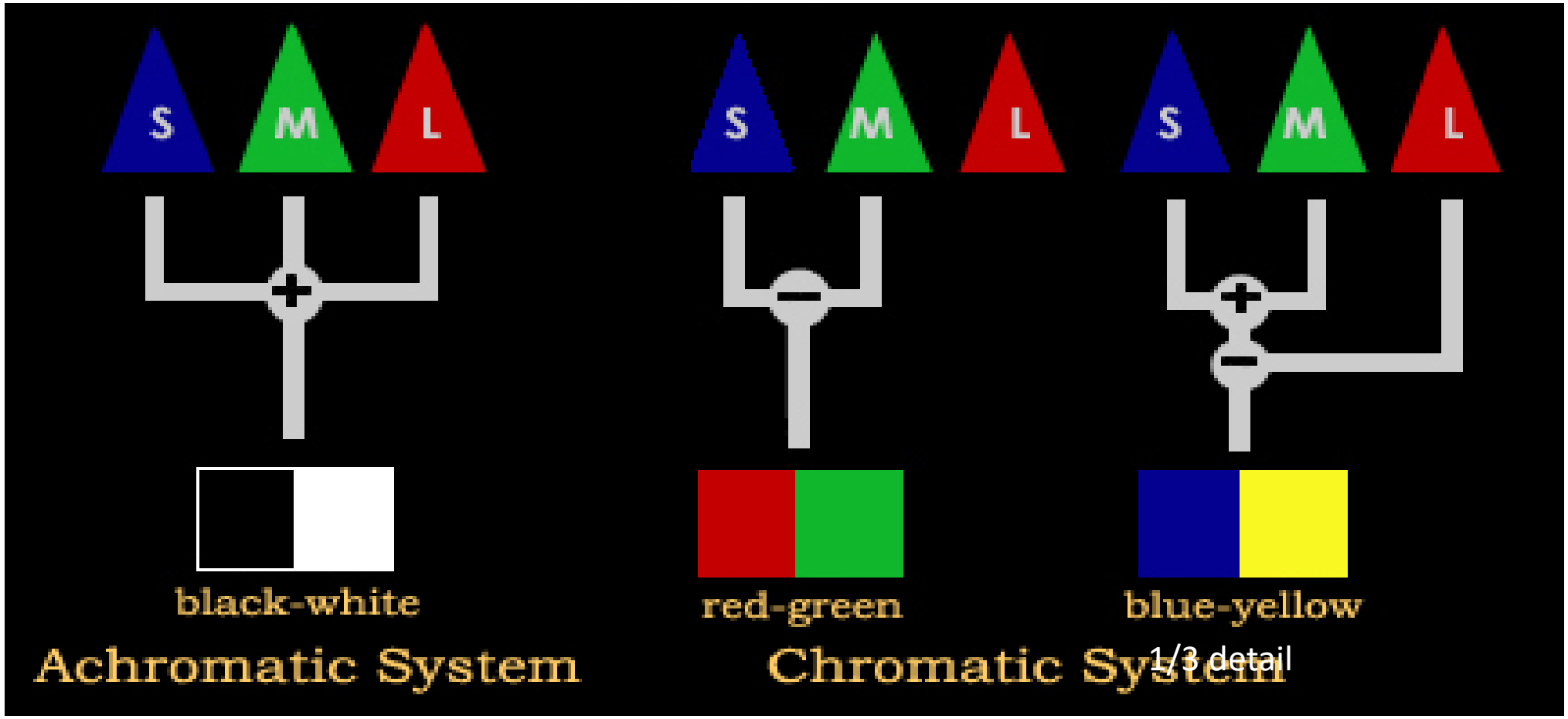
Color spaces and perceptual uniformity







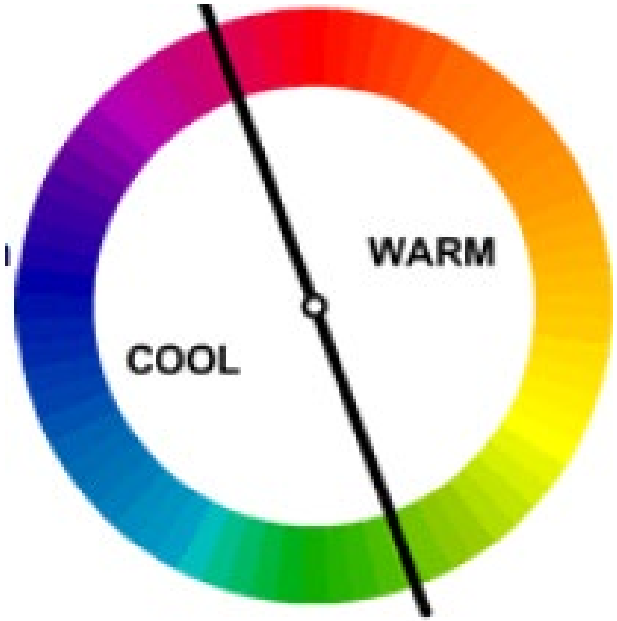
The opponent-process theory (Edwald Hering)



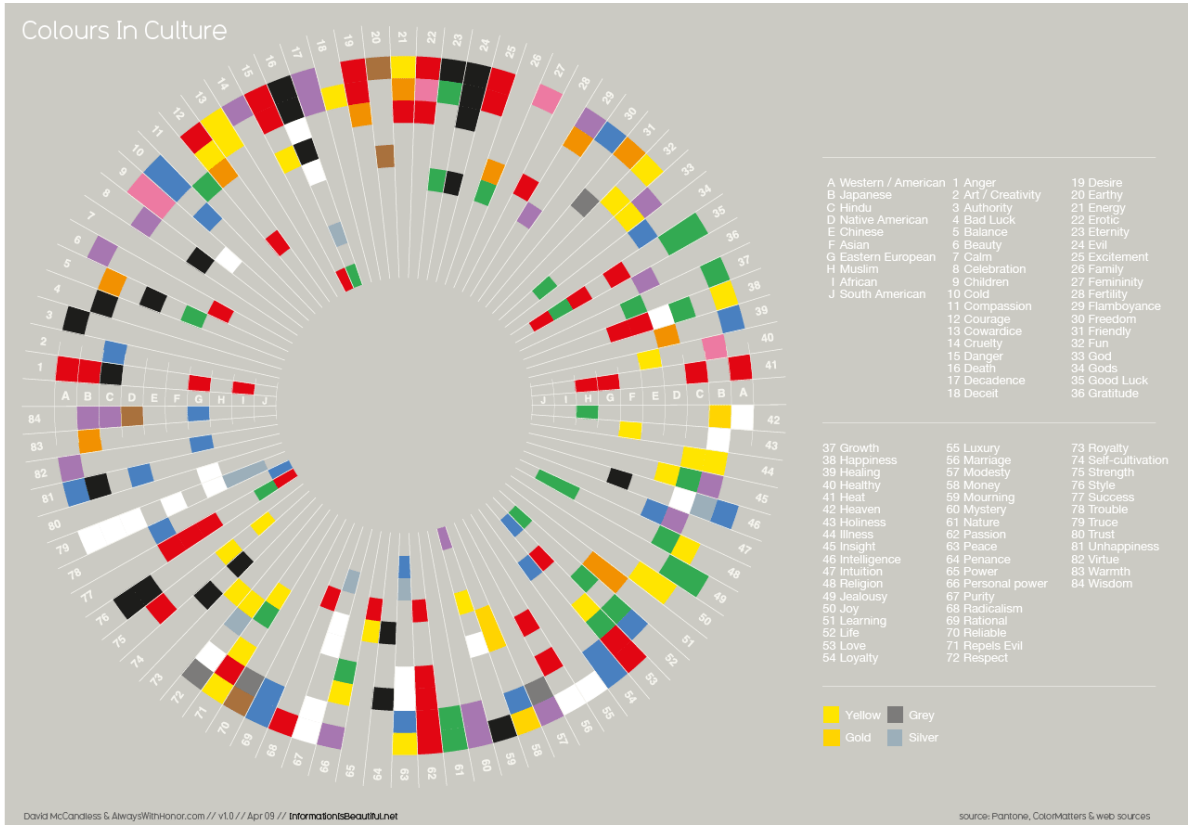
1/3 detail

Colour affect our moods

- Colours affect us in numerous ways, both mentally and physically
- We can divide colours into warm and cool
  - Warm colours are energetic and tend to advance in space.
  - Cool colours give calm and tend to work better as background.



Colour meaning is a cultural issue



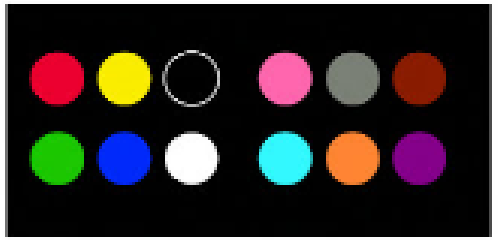
Color for charts

- Assign colour according to function:
  - Use contrast to highlight
  - Analogous colours to group
  - Use greys for context and axis when labelling with colour

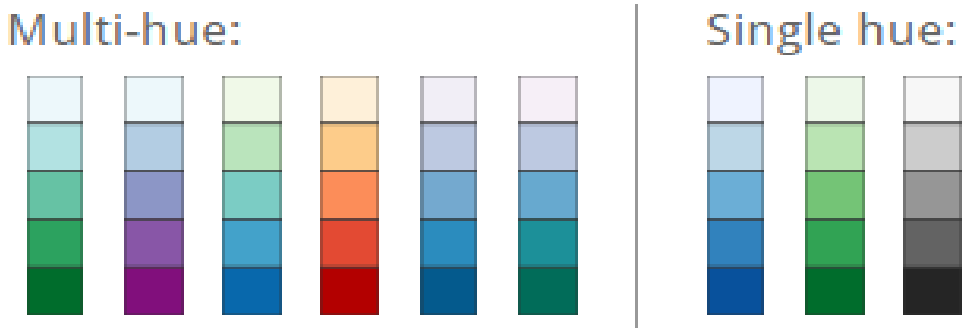
Data types and colour

Categorical data





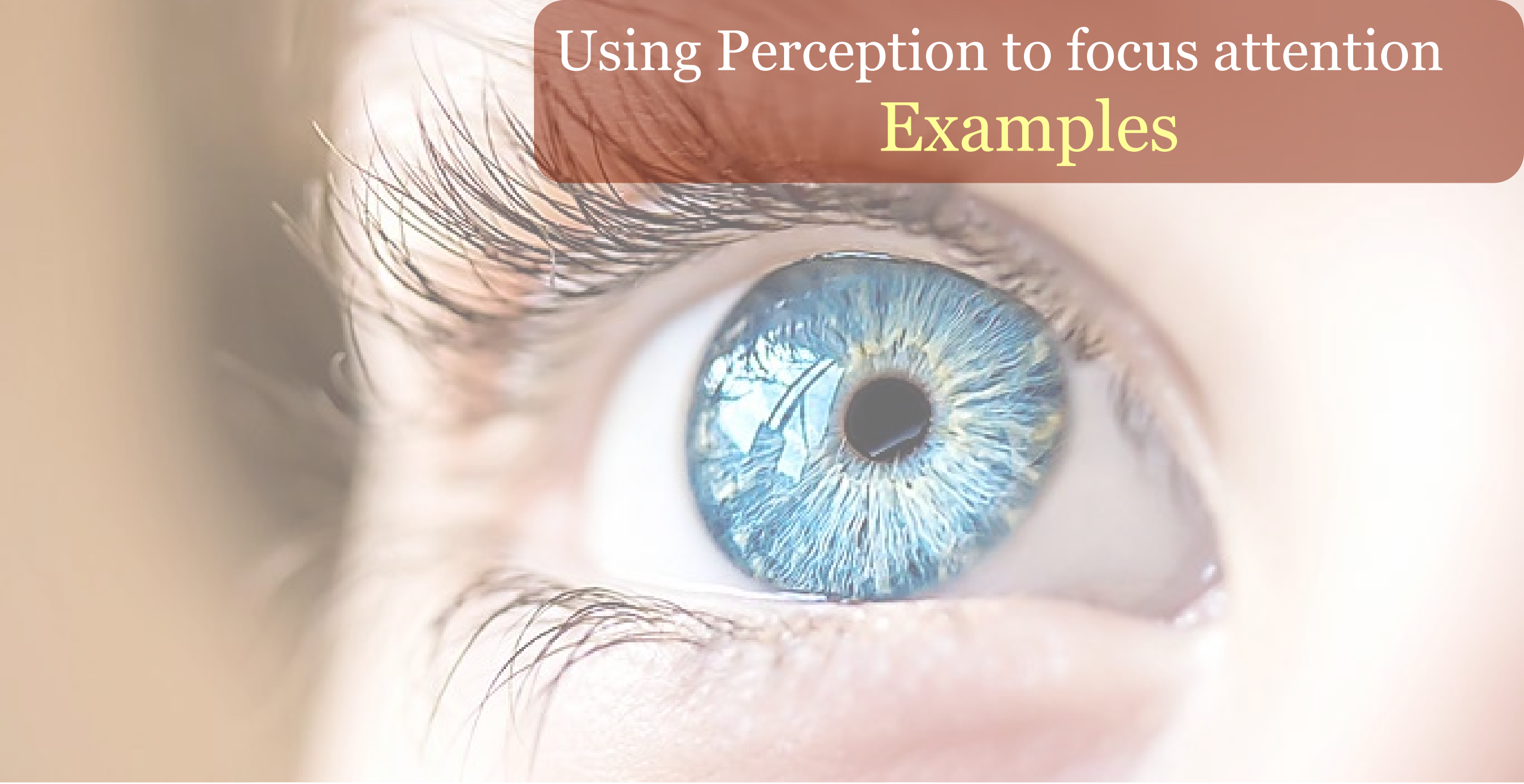
Numerical data



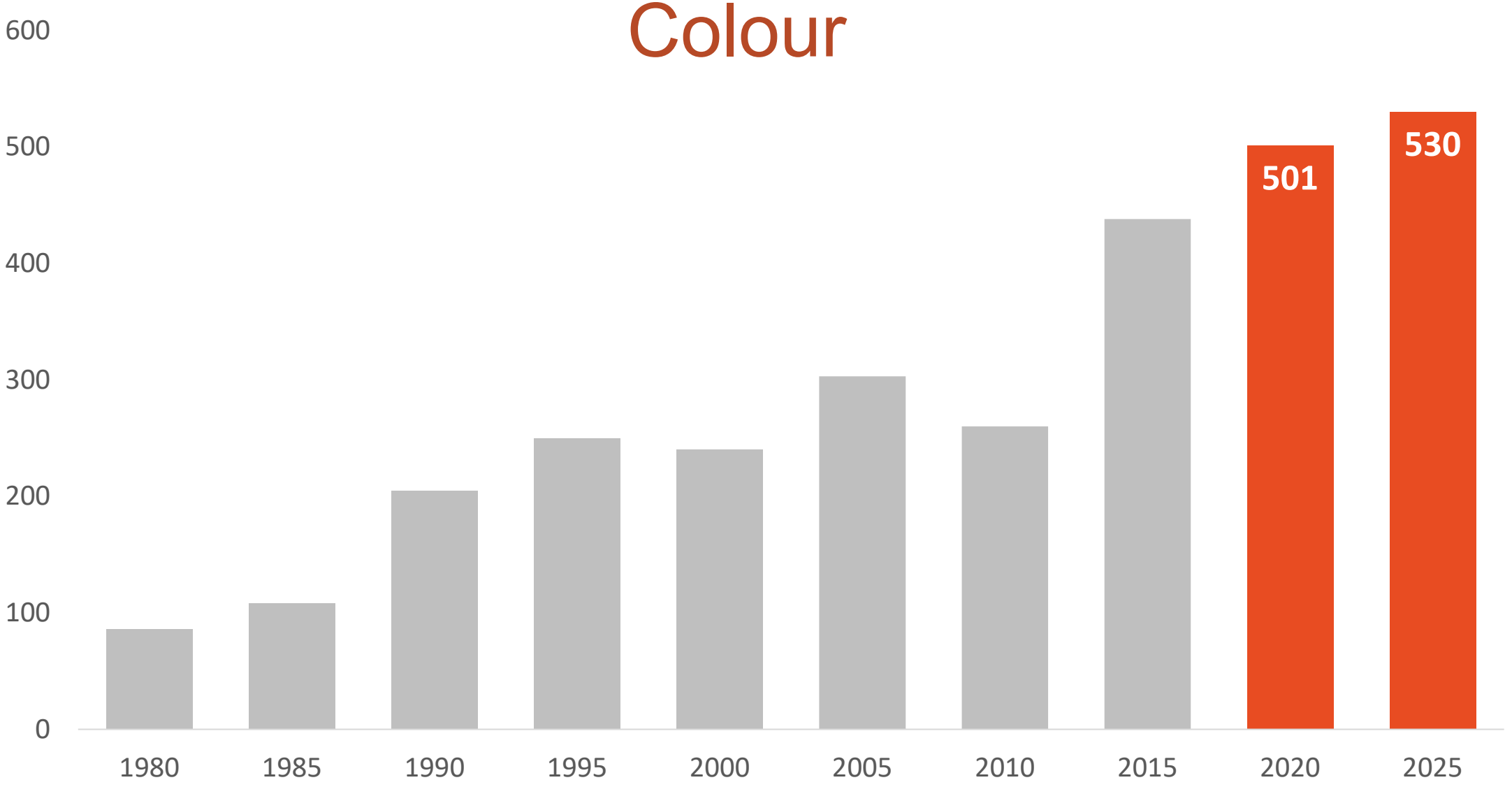


Using Perception to focus attention

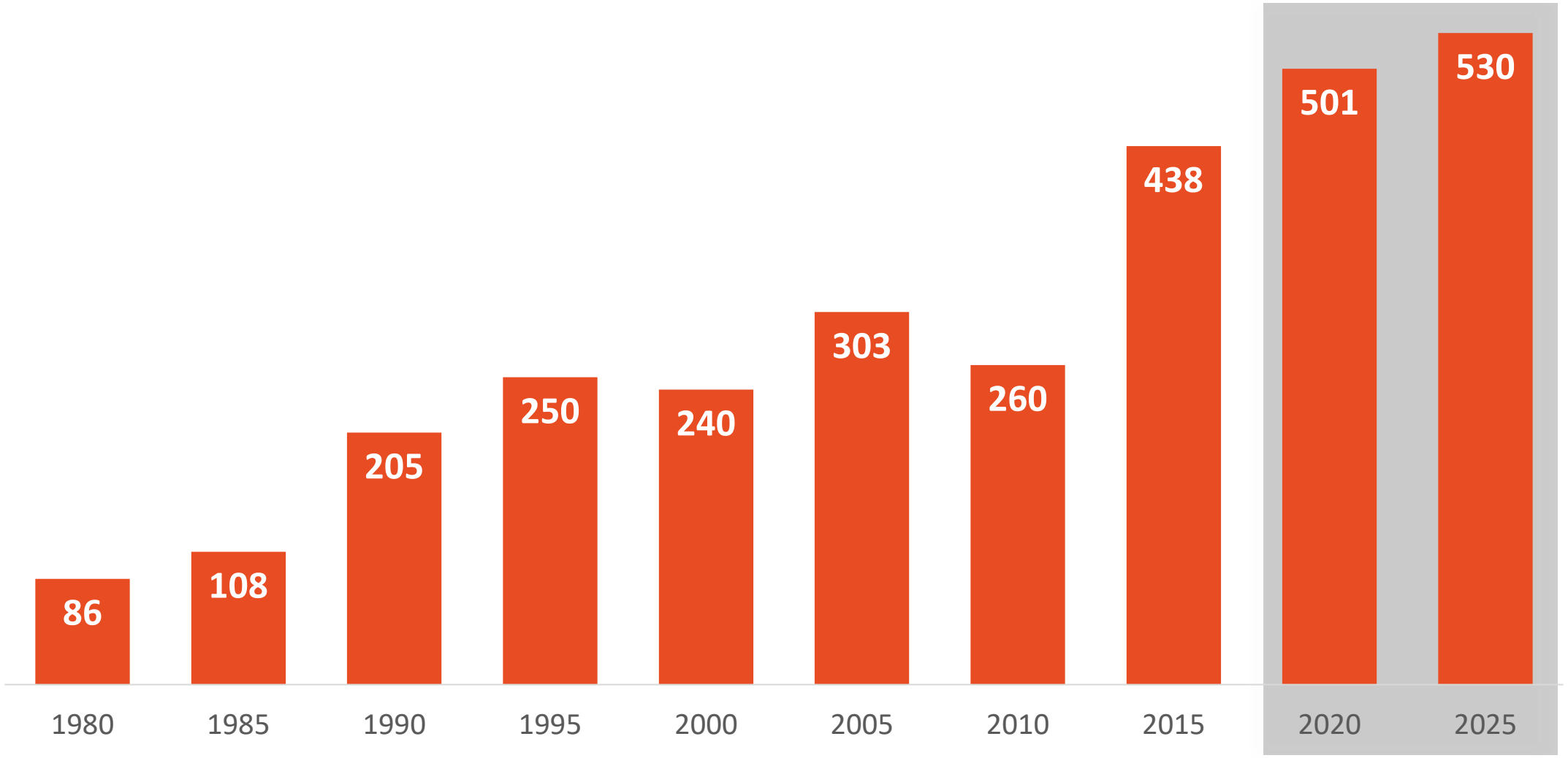
Examples



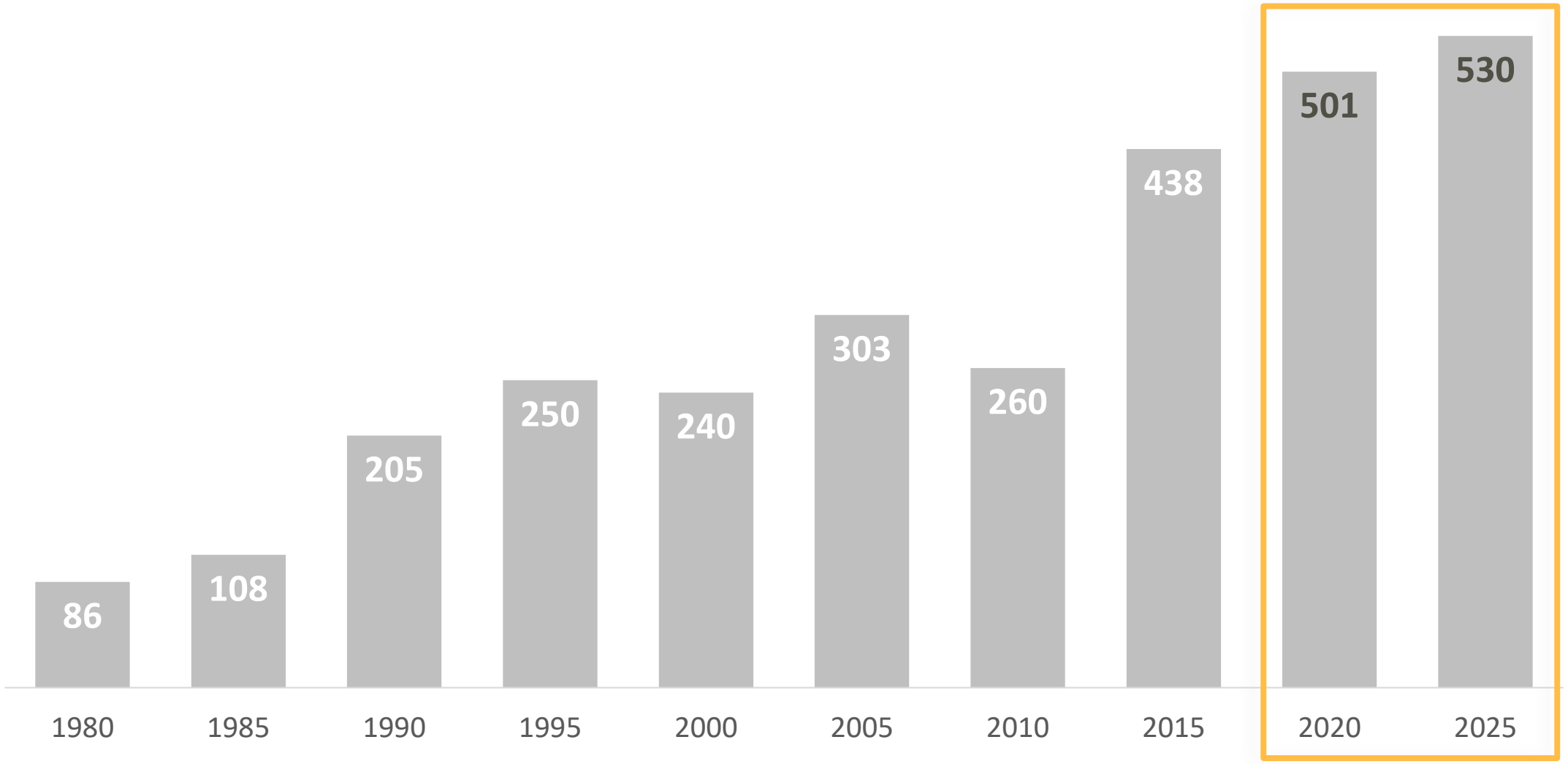
Colour


600
500
400
300
200
100
0
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
501
530

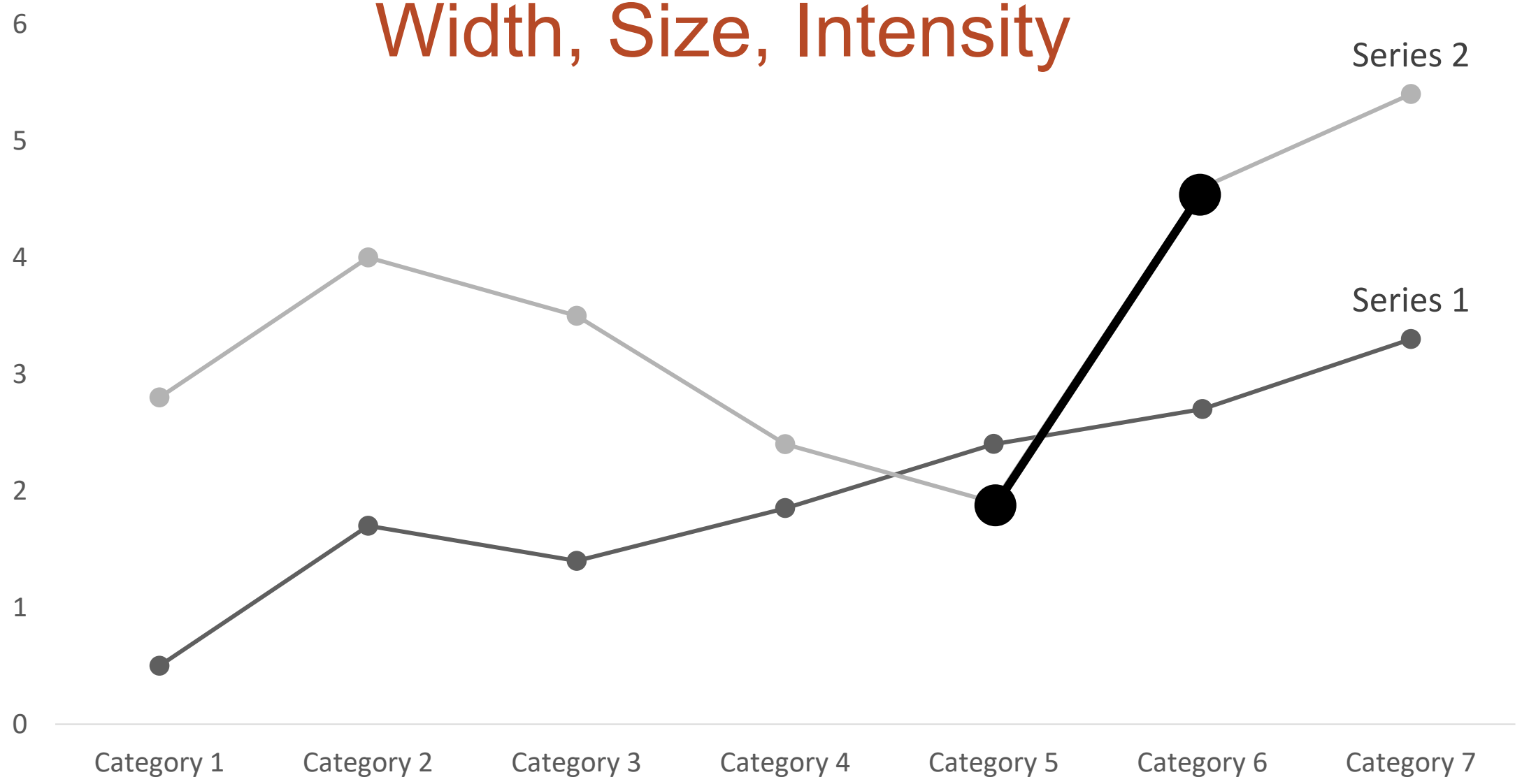
Enclosure


86
108
205
250
240
303
260
438
501
530
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025

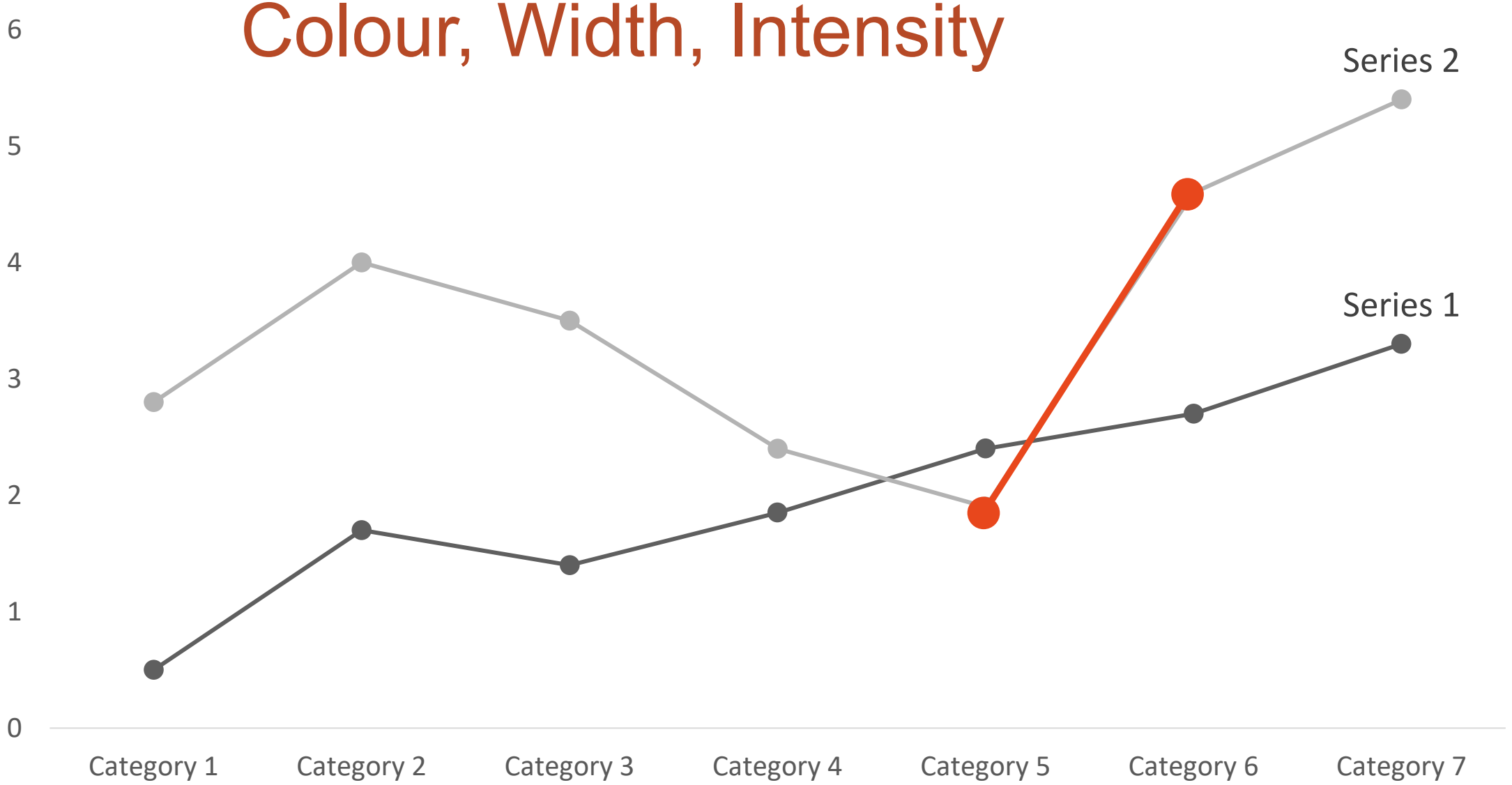
Enclosure


86
108
205
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1980
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2020
2025

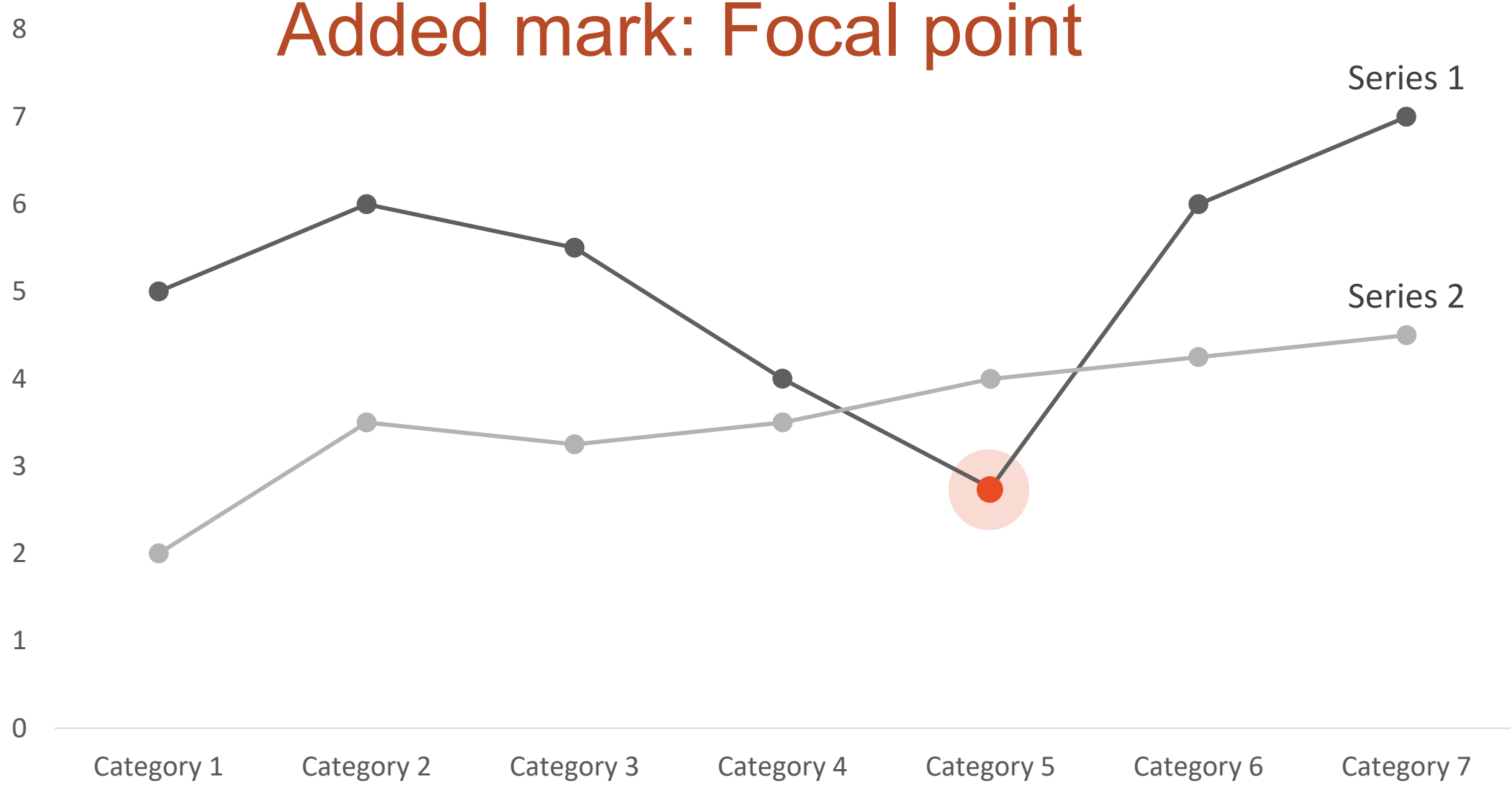
Width, Size, Intensity



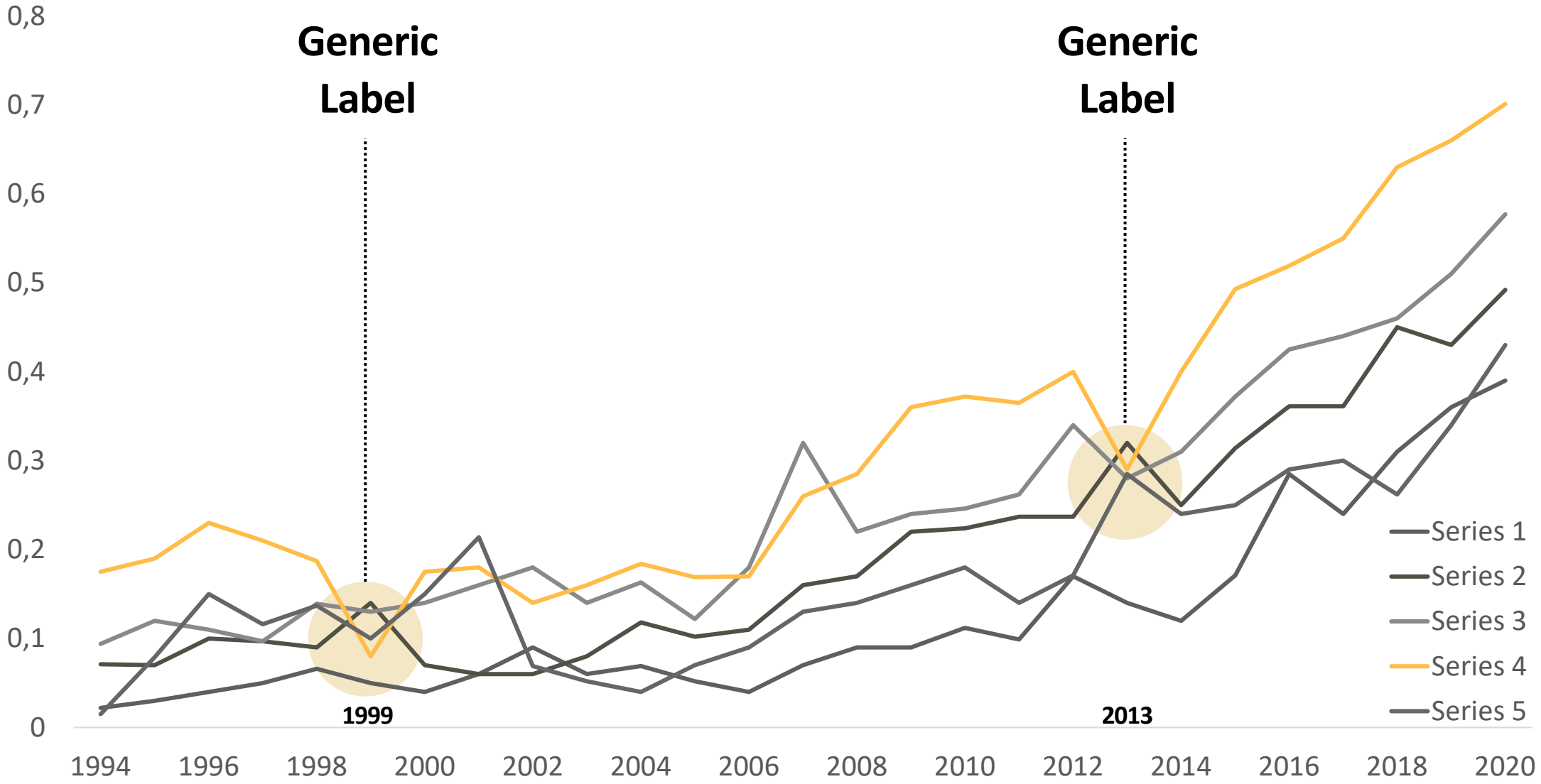
Colour, Width, Intensity



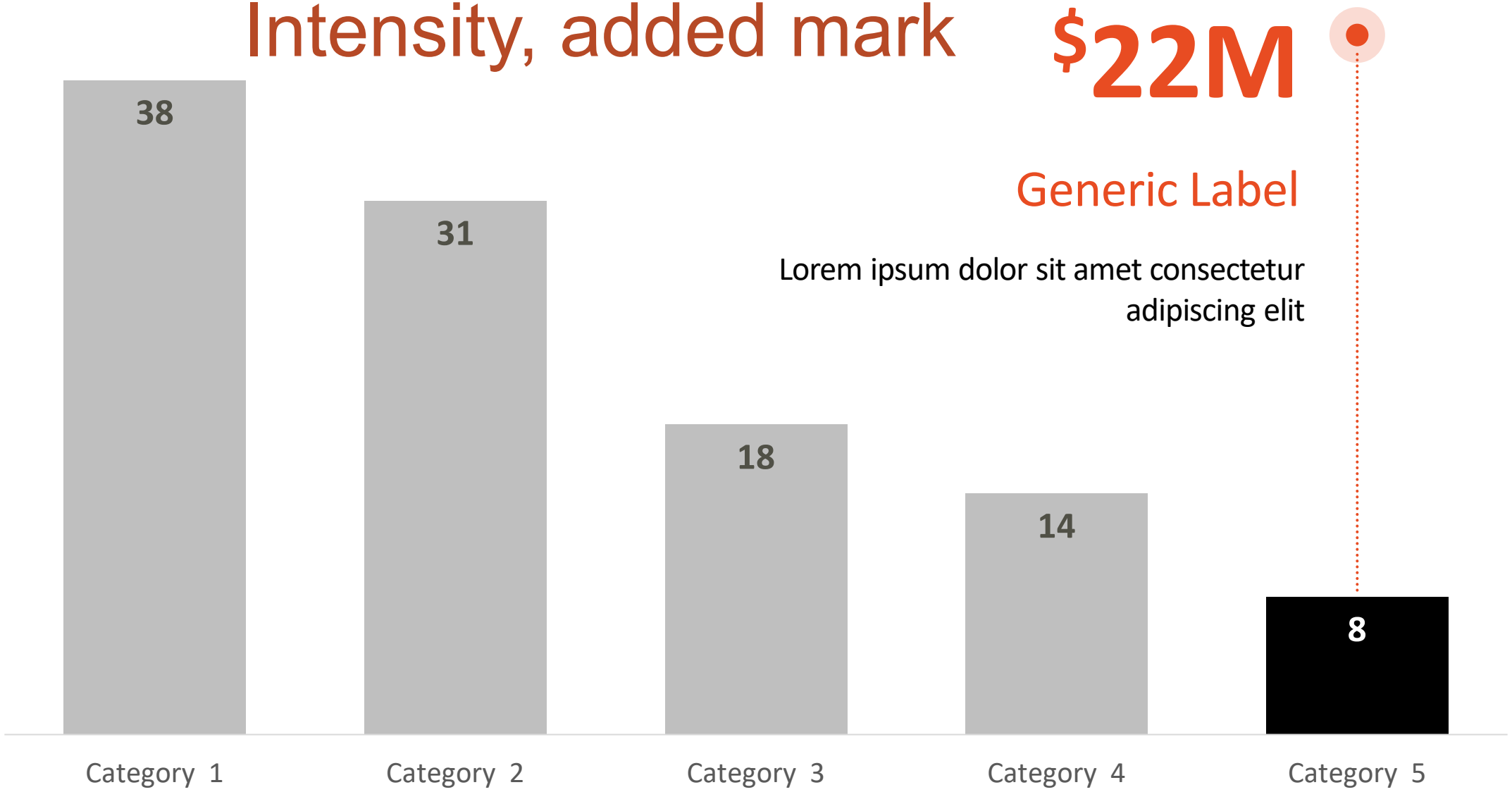
Added mark: Focal point



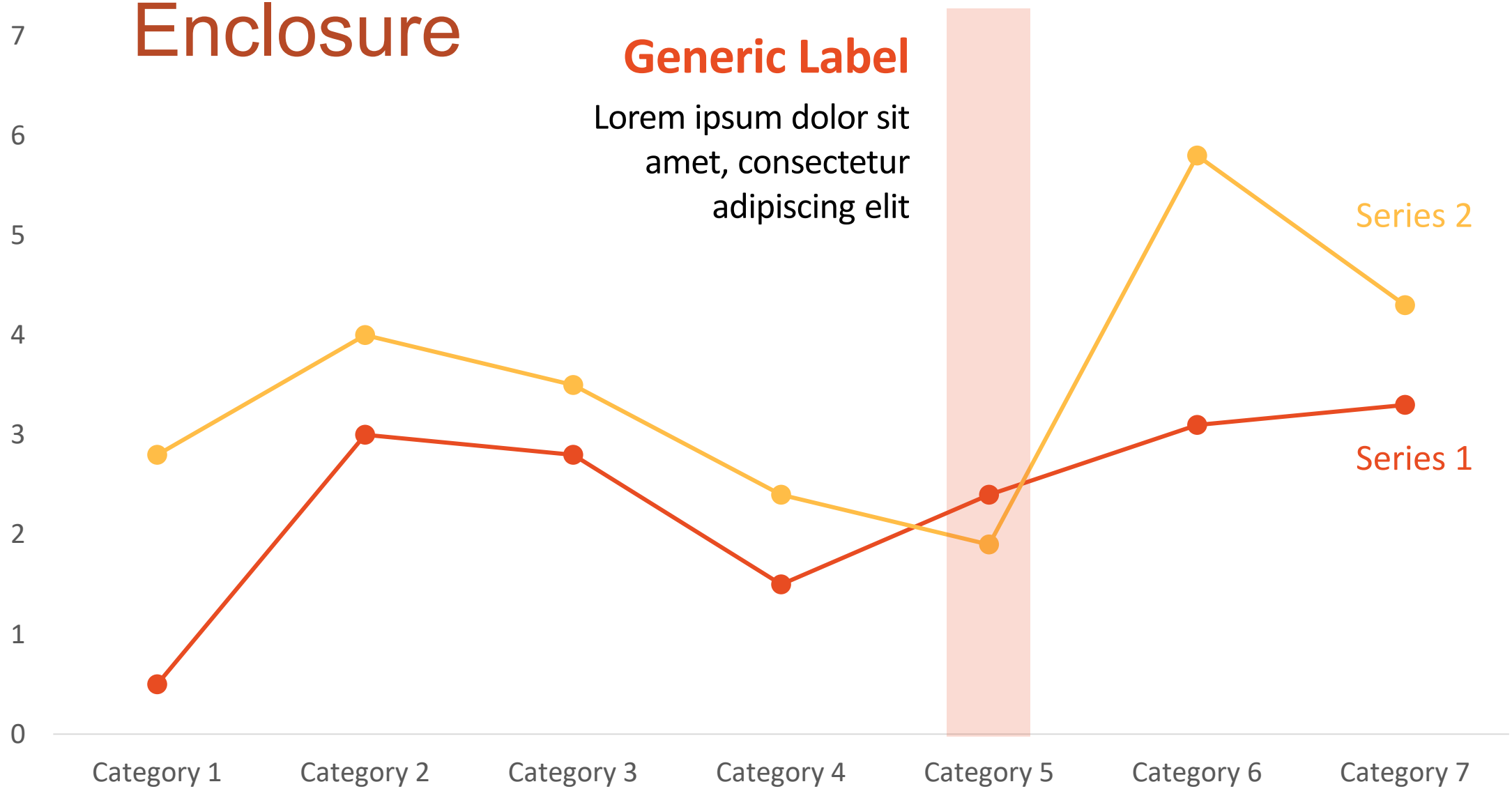
Added mark: Focal point



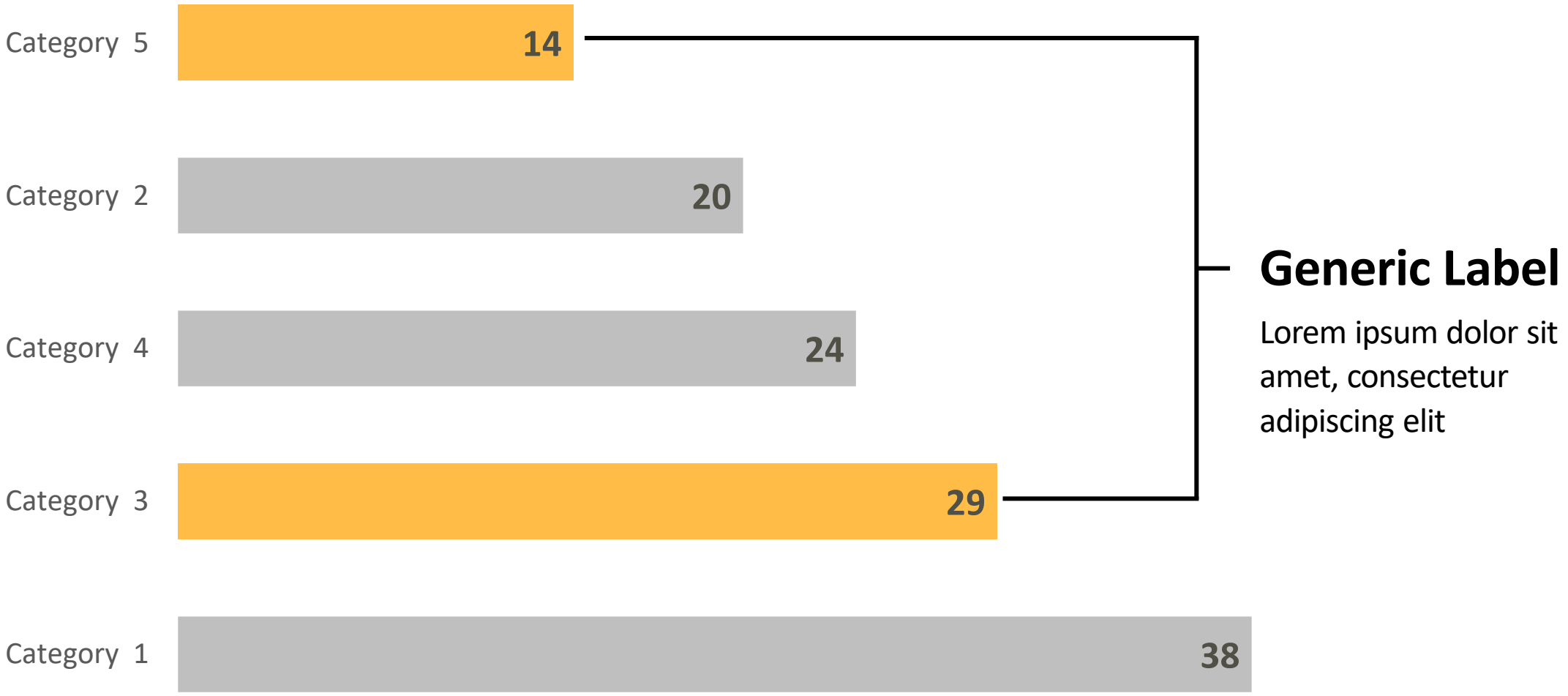
Intensity, added mark



Enclosure


Generic Label
Lorem ipsum dolor sit amet, consectetur adipiscing elit
7
6
5
4
3
2
1
0
Category 1
Category 2
Category 3
Category 4
Category 5
Category 6
Category 7
Series 2
Series 1

Connectedness, similarity



Combining preattentive properties + gestalt to represent quantity

- size:
  - *length or height,*
  - *area* (radius),
  - never *volume*
- lightness, darker = bigger
- hue saturation, saturated = bigger
- vertical position, higher = bigger

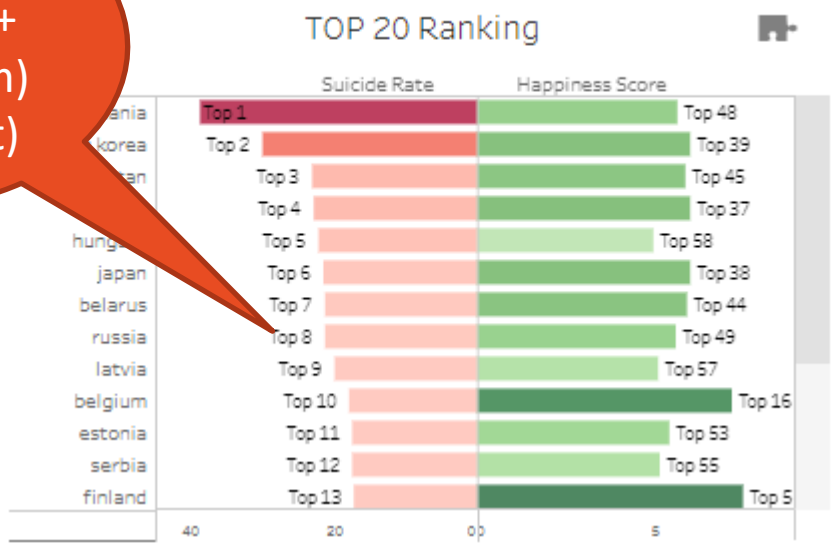
Combining preattentive properties + gestalt to represent INTENSITY

- Darker or more saturated,
- Bigger,
- Thicker

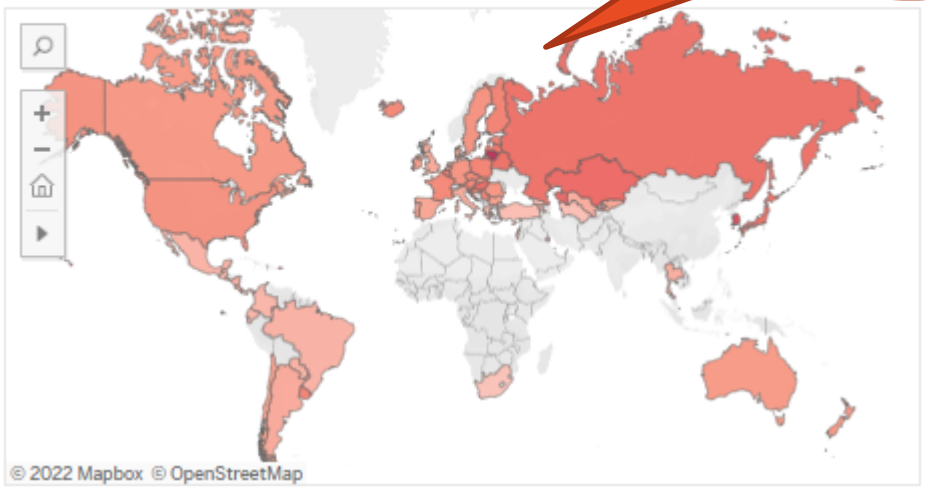
Combining preattentive properties + gestalt to represent VISUAL SALIENCE

- Distinct from the norm: in hue, orientation,
- Enclosure: by line or background colour,
- Added marks

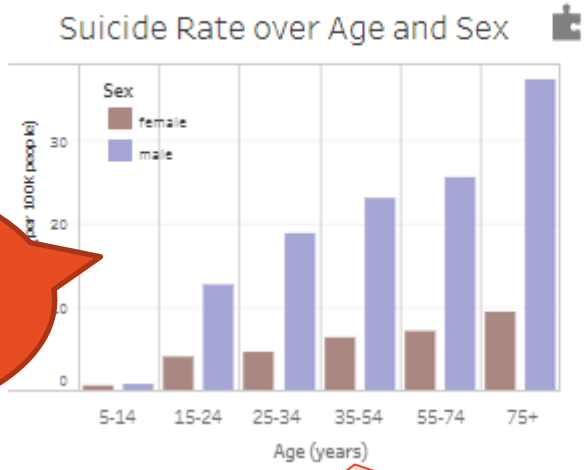
Color (Hue + Intensity) + length(Form) (redundant)

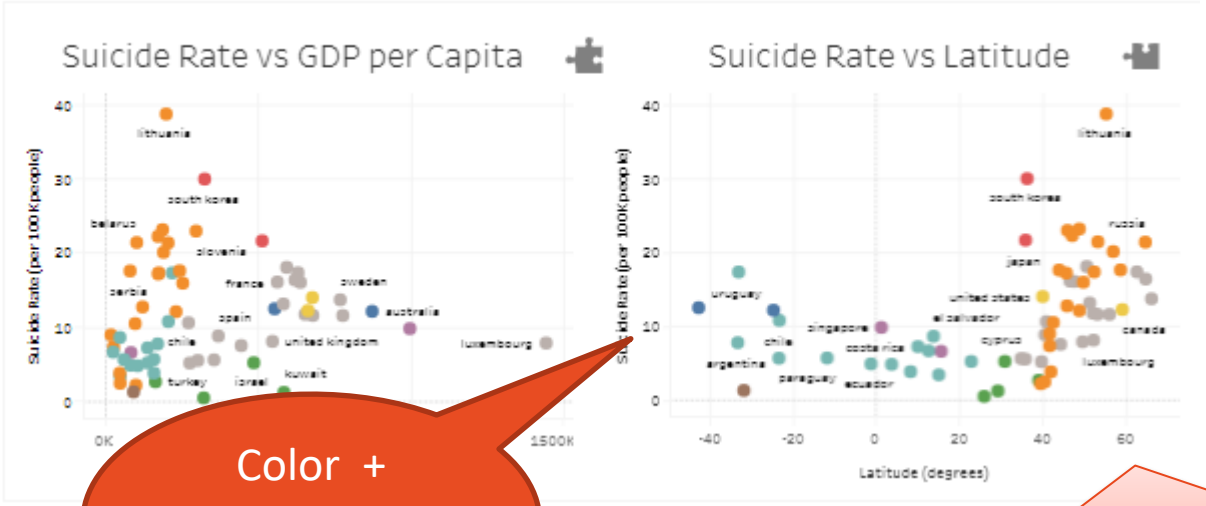


Color (Saturation + Luminosity) = Intensity



Color + length (Form) (separable)





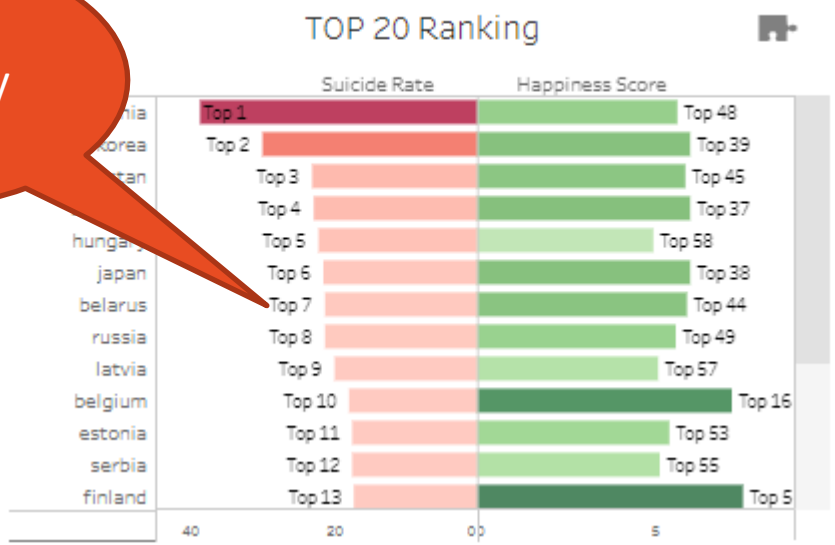
Color + Position (separable)

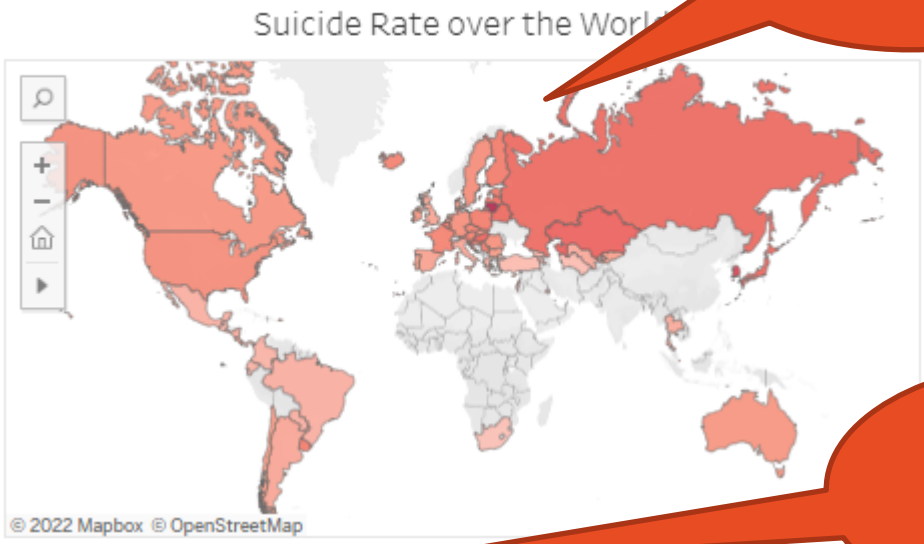
What do women…? What age …?

How is Central and Eastern Europe GDP and suicide rate?

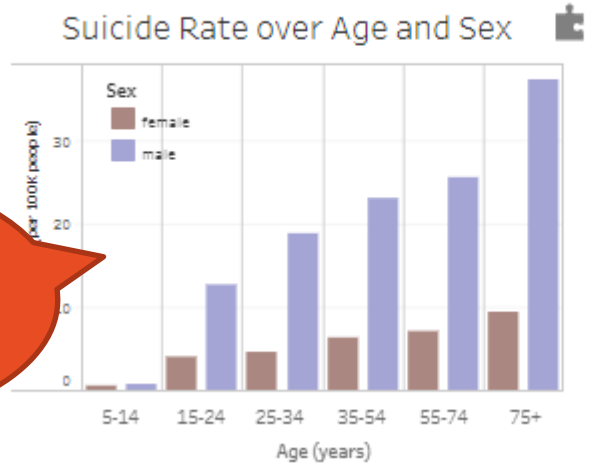
Symmetry

Similarity



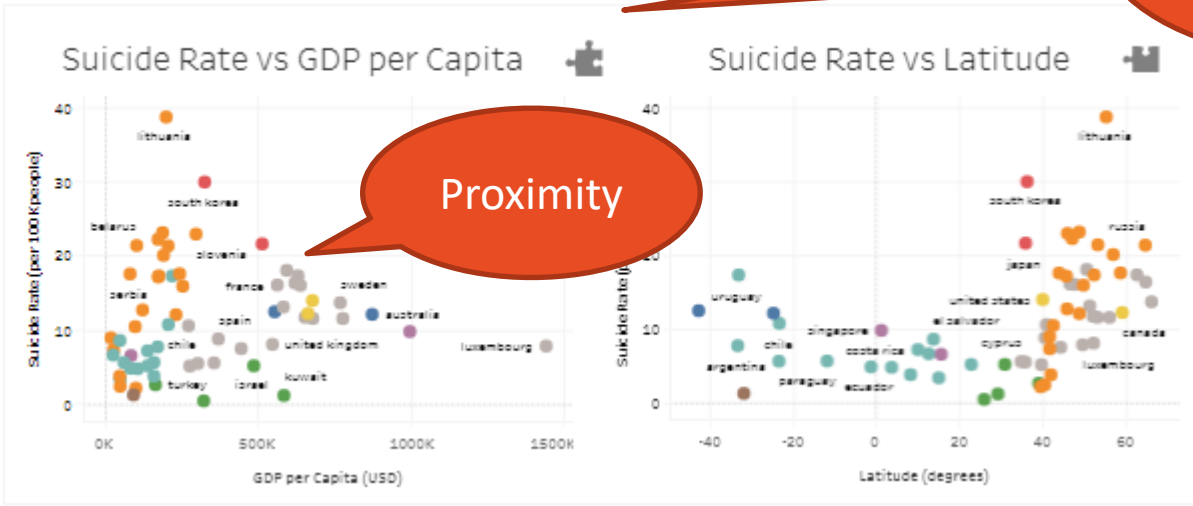


Closure

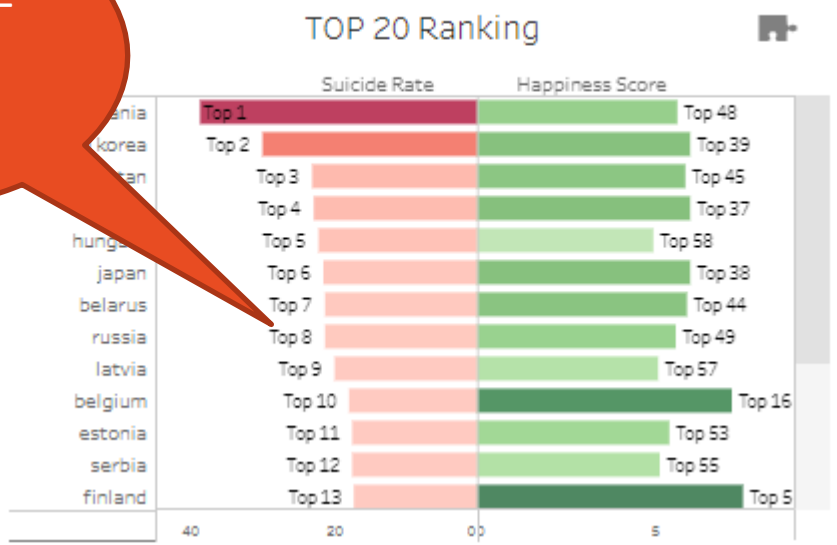


Similarity

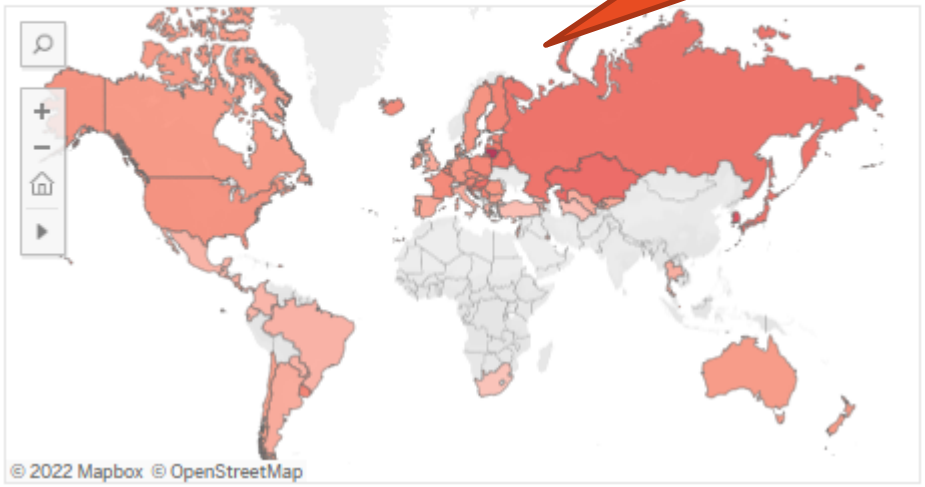
Proximity



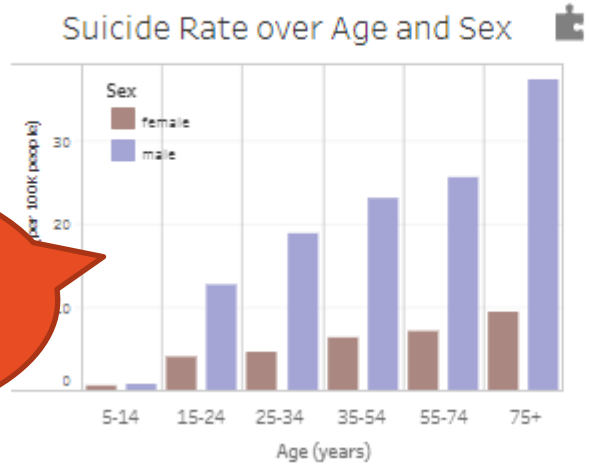
More saturated = bigger
Larger = bigger



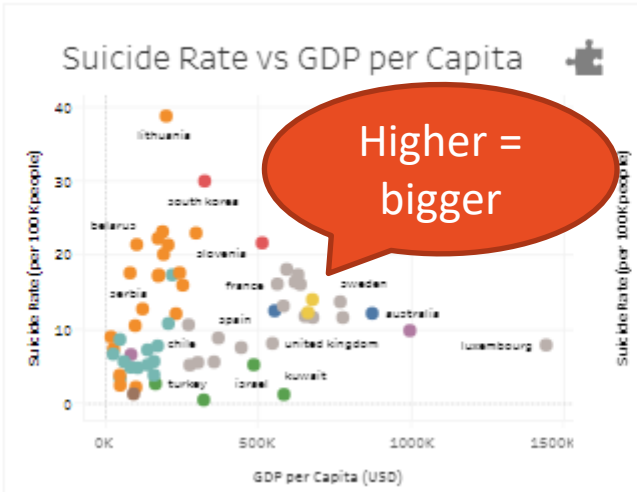
More saturated = bigger

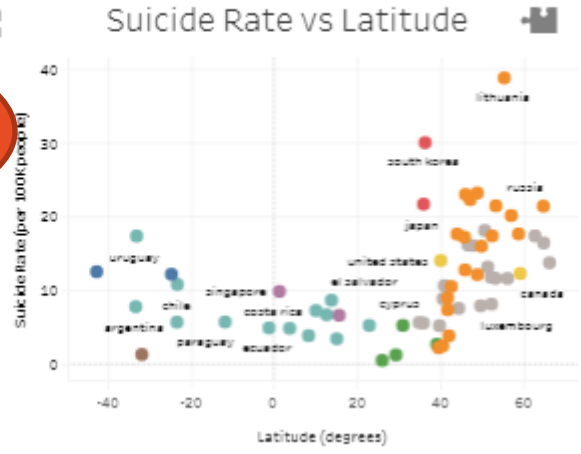


Larger = bigger



Higher = bigger





Sources

- Ware, C. (2020) Information visualization: perception for design. Burlington: Morgan Kaufmann.
- Heeger, D (2006) Perception Lecture Notes: Retinal Ganglion Cells. NY: Department of Psychology, New York University.
- Healey, CG (2012) Perception in Visualization. Department of Computer Science, North Carolina State University.
- Few, S (2012) Show Me the Numbers: Designing Tables and Graphs to Enlighten. Analytics Press.
- van Dijk, D. (2022) Mind your data visualization. Datylon BV.
- Satyanarayand, Arvin (2023) Graphical perception

Thank you for your attention





UNIVERSITAT DE BARCELONA

Key ideas

1. Information visualization is a tool that uses our visual perception capacity to digest data and facilitate its understanding.

2. Preattentive properties allow maximum efficiency in data communication. Gestalt laws guide perception.

3. By combining both we can highlight and relate specific graphs and data.